

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275650

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Ciccone; John R. et al.

SPACE EFFICIENT GAS VALVES FOR GRILLS

Abstract

Space efficient gas valves for grills are disclosed. An example gas valve includes a chamber, a flow control member, an inlet conduit, an outlet conduit, and a stem. The flow control member is disposed within the chamber and is rotatable between an open position and a closed position. The inlet conduit extends downwardly from the chamber and includes an inlet flow channel having an inlet. The outlet conduit extends upwardly from the chamber and includes an outlet flow channel having an outlet. The stem is operatively coupled to the flow control member such that rotation of the stem about an axis of rotation of the stem causes a corresponding rotation of the flow control member within the chamber about an axis of rotation of the flow control member. The outlet of the outlet flow channel is located above the axis of rotation of the of the flow control member.

Inventors: Ciccone; John R. (Barrington, IL), Lee; Joshua (Palatine, IL), Szulakiewicz; Sebastian (Mount Prospect, IL)

Applicant: Weber-Stephen Products LLC (Palatine, IL)

Family ID: 96881349

Appl. No.: 18/595362

Filed: March 04, 2024

Publication Classification

Int. Cl.: A47J37/07 (20060101); F23K5/00 (20060101)

U.S. Cl.:

CPC A47J37/0786 (20130101); F23K5/007 (20130101); F23K2900/05002 (20130101)

Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure relates generally to gas valves and, more specifically, to space efficient gas valves for grills.

BACKGROUND

[0002] Gas grills are typically equipped with a burner assembly including a manifold, a burner tube, and a gas valve (e.g., a control valve), with the gas valve being operatively positioned between the manifold and the burner tube to control a flow of pressurized fluid (e.g., pressurized gas) from the manifold into the gas valve, and from the gas valve into the burner tube. In such conventional gas grill implementations, it is common for the outlet of the gas valve leading into the burner tube to be coaxially aligned with an axis of rotation of a flow control member of the gas valve, and/or to be coaxially aligned with an axis of rotation of a stem that controls the rotation of such a flow control member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a first perspective view of an example gas valve constructed in accordance with the teachings of this disclosure.

[0004] FIG. 2 is a second perspective view of the gas valve of FIG. 1.

[0005] FIG. 3 is a right side view of the gas valve of FIGS. 1 and 2.

[0006] FIG. 4 is a left side view of the gas valve of FIGS. 1-3.

[0007] FIG. 5 is a front view of the gas valve of FIGS. 1-4.

[0008] FIG. 6 is a rear view of the gas valve of FIGS. 1-5.

[0009] FIG. 7 is a top view of the gas valve of FIGS. 1-6.

[0010] FIG. 8 is a bottom view of the gas valve of FIGS. 1-7.

[0011] FIG. 9 is a partial cutaway view of the gas valve of FIGS. 1-8 taken along section A-A of FIG. 5, with the flow control member of the gas valve omitted for enhanced viewability.

[0012] FIG. 10 is a cross-sectional view of the gas valve of FIGS. 1-9 taken along section A-A of FIG. 5, with the flow control member of the gas valve omitted for enhanced viewability.

[0013] FIG. 11 is a partial cutaway view of the gas valve of FIGS. 1-10 taken along section B-B of FIG. 7, with the flow control member of the gas valve omitted for enhanced viewability.

[0014] FIG. 12 is a first perspective view of an example cone constructed in accordance with the teachings of this disclosure.

[0015] FIG. 13 is a second perspective view of the cone of FIG. 12.

[0016] FIG. 14 is a right side view of the cone of FIGS. 12 and 13.

[0017] FIG. 15 is a left side view of the cone of FIGS. 12-14.

[0018] FIG. 16 is a front view of the cone of FIGS. 12-15.

[0019] FIG. 17 is a rear view of the cone of FIGS. 12-16.

[0020] FIG. 18 is a top view of the cone of FIGS. 12-17.

[0021] FIG. 19 is a bottom view of the cone of FIGS. 12-18.

[0022] FIG. 20 is a partial cutaway view taken along section A-A of FIG. 5, showing the cone of FIGS. 12-19 disposed in the chamber of the gas valve of FIGS. 1-11, with the cone in an example closed position.

[0023] FIG. 21 is a cross-sectional view taken along section A-A of FIG. 5, showing the cone of FIGS. 12-20 disposed in the chamber of the gas valve of FIGS. 1-11 and 20, with the cone in the closed position of FIG. 20.

[0024] FIG. 22 is a partial cutaway view taken along section B-B of FIG. 7, showing the cone of FIGS. 12-21 disposed in the chamber of the gas valve of FIGS. 1-11, 20, and 21, with the cone in the closed position of FIGS. 20 and 21.

[0025] FIG. 23 is a partial cutaway view taken along section A-A of FIG. 5, showing the cone of FIGS. 12-22 disposed in the chamber of the gas valve of FIGS. 1-11 and 20-22, with the cone in an example open position.

[0026] FIG. 24 is a cross-sectional view taken along section A-A of FIG. 5, showing the cone of FIGS. 12-23 disposed in the chamber of the gas valve of FIGS. 1-11 and 20-23, with the cone in the open position of FIG. 23.

[0027] FIG. 25 is a partial cutaway view taken along section B-B of FIG. 7, showing the cone of FIGS. 12-24 disposed in the chamber of the gas valve of FIGS. 1-11 and 20-24, with the cone in the open position 2300 of FIGS. 23 and 24.

[0028] FIG. 26 is a perspective view of an example grill constructed in accordance with the teachings of this disclosure.

[0029] FIG. 27 is an exploded view of the grill of FIG. 26.

[0030] FIG. 28 is a perspective view showing the first burner tube of the grill of FIGS. 26 and 27 in isolation.

[0031] FIG. 29 is a top view of the first burner tube of FIG. 28.

[0032] FIG. 30 is a bottom view of the first burner tube of FIGS. 28 and 29.

[0033] FIG. 31 is a right side view of the first burner tube of FIGS. 28-30.

[0034] FIG. 32 is a cross-sectional view of the first burner tube of FIGS. 28-31 taken along section C-C of FIG. 29.

[0035] FIG. 33 is a perspective view of a portion of an example burner assembly of the grill of FIGS. 26 and 27.

[0036] FIG. 34 is a side view of the burner assembly of FIG. 33.

[0037] FIG. 35 is a front view showing the grill of FIGS. 26 and 27, with the frame, the lid, and certain other structural features of the grill omitted for enhanced viewability.

[0038] FIG. 36 is a top view of the grill of FIG. 35.

[0039] FIG. 37 is a cross-sectional view of the grill of FIGS. 35 and 36 taken along section D-D of FIG. 36.

[0040] FIG. 38 is a cross-sectional view of the grill of FIGS. 35-37 taken along section E-E of FIG. 35.

[0041] Certain examples are shown in the above-identified figures and described in detail below. In describing these examples, like or identical reference numbers are used to identify the same or similar elements. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale or in schematic for clarity and/or conciseness.

[0042] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name.

DETAILED DESCRIPTION

[0043] As discussed above, gas grills are typically equipped with a burner assembly including a manifold, a burner tube, and a gas valve (e.g., a control valve), with the gas valve being operatively positioned between the manifold and the burner tube to control a flow of pressurized fluid (e.g., pressurized gas) from the manifold into the gas valve, and from the gas valve into the burner tube. In such conventional gas grill implementations, it is common for the outlet of the gas valve leading into the burner tube to be coaxially aligned with an axis of rotation of a flow control member of the gas valve, and/or to be coaxially aligned with an axis of rotation of a stem that controls the rotation of such a flow control member.

[0044] The aforementioned relationship between parts results in a gas valve that is spatially inefficient, particularly with regard to the front-to-rear distance between the stem of the gas valve and the burner tube of the gas grill. As a result, conventional gas grills often include a control panel chamber (e.g., the area between a control panel of the gas grill and a front wall of a cookbox of the gas grill) that is undesirably large and/or bulky, particularly along the front-to-rear dimension of the control panel chamber.

[0045] Example gas valves disclosed herein provide significant improvements with regard to spatial efficiency relative to the gas valves of the conventional gas grills described above. In some disclosed examples, a gas valve includes a chamber, a flow control member, an inlet conduit, an outlet conduit, and a stem. The flow control member is disposed within the chamber. The flow control member is rotatable within the chamber between an open position and a closed position. The inlet conduit extends downwardly from the chamber. The inlet conduit includes an inlet flow channel having an inlet. The outlet conduit extends upwardly from the chamber. The outlet conduit includes an outlet flow channel having an outlet. The stem is operatively coupled to the flow control member such that rotation of the stem about an axis of rotation of the stem causes a corresponding rotation of the flow control member within the chamber about an axis of rotation of the flow control member. The outlet of the outlet flow channel is located above the axis of rotation of the flow control member.

[0046] In some disclosed examples, the inlet conduit is configured to be coupled to a manifold of a grill such that the inlet is in fluid communication with the manifold, and the outlet conduit is configured to be coupled to a burner tube of the grill such that the outlet is in fluid communication with the burner tube.

[0047] In some disclosed examples, the axis of rotation of the flow control member is coaxially aligned with the axis of rotation of the stem.

[0048] In some disclosed examples, the chamber includes an inlet opening and an outlet opening, and the flow control member includes a flow chamber having an inlet opening and an outlet opening. In some disclosed examples, the inlet flow channel of the inlet conduit is in fluid communication with the flow chamber of the flow control member via the inlet opening of the chamber and the inlet opening of the flow chamber when the flow control member is in the open position, and the inlet flow channel of the inlet conduit is not in fluid communication with the flow chamber of the flow control member when the flow control member is in the closed position. In some disclosed examples, the outlet flow channel of the outlet conduit is in fluid communication with the flow chamber of the flow control member via the outlet opening of the flow control member and the outlet opening of the chamber when the flow control member is in the open position, and the outlet flow channel of the outlet conduit remains in fluid communication with the flow chamber of the flow control member via the outlet opening of the flow control member and the outlet opening of the chamber when the flow control member is in the closed position.

[0049] In some disclosed examples, the outlet opening of the chamber is a first outlet opening of the chamber and the chamber further includes a second outlet opening, the outlet opening of the flow chamber is a first outlet opening of the flow chamber and the flow chamber further includes a second outlet opening, the outlet conduit is a first outlet conduit of the gas valve, the outlet flow channel is a first outlet flow channel, and the outlet is a first outlet, and the gas valve further includes a second outlet conduit including a second outlet flow channel having a second outlet. In some disclosed examples, the second outlet flow channel of the second outlet conduit is in fluid communication with the flow chamber of the flow control member via the second outlet opening of the flow chamber and the second outlet opening of the chamber when the flow control member is in the open position, and the second outlet flow channel of the second outlet conduit is not in fluid communication with the flow chamber of the flow control member when the flow control member is in the closed position. In some disclosed examples, the second outlet conduit is configured to be coupled to an ignition conduit of a grill such that the second outlet is in fluid communication with

the ignition conduit.

[0050] In some disclosed examples, a gas train of the gas valve includes an S-shaped portion defined by the chamber and the outlet flow channel. In some disclosed examples, the outlet flow channel includes a first segment extending upwardly from and being in fluid communication with the chamber, a second segment extending forwardly from and being in fluid communication with the first segment, a third segment extending upwardly from and being in fluid communication with the second segment, and a fourth segment extending rearwardly from and being in fluid communication with the third segment. In some disclosed examples, the fourth segment includes the outlet, and the S-shaped portion of the gas train includes the chamber, the first segment, the second segment, the third segment, and the fourth segment. In some disclosed examples, the first segment, the second segment, the third segment, and the fourth segment are linear. In some disclosed examples, a central axis of the chamber, a central axis of the second segment, and a central axis of the fourth segment are parallel to one another and horizontally oriented, and a central axis of the first segment and a central axis of the third segment are parallel to one another and vertically oriented. In some disclosed examples, the central axis of the second segment is located above the central axis of the chamber, the central axis of the fourth segment is located above the central axis of the second segment, the central axis of the third segment is located forward of the central axis of the first segment, and the central axis of the third segment is parallel to and located forward of a central axis of the inlet flow channel.

[0051] The above-described features of the disclosed gas valves advantageously provide significant improvements with regard to spatial efficiency relative to gas valves of the conventional gas grills described above. In this regard, positioning the outlet of the gas valve at a location that is offset from (e.g., located above) the axis of rotation of the flow control member of the gas valve, and/or offset from (e.g., located above) the axis of rotation of the stem of the gas valve enables the front-to-rear distance between the stem of the gas valve and the outlet of the gas valve to be reduced (e.g., minimized). The improved space efficiency associated with this reduced dimension of the gas valve advantageously enables the gas valve to be disposed within a correspondingly smaller (e.g., reduced depth) control panel chamber of a grill, and also advantageously enables the burner ports of a burner tube of such a grill to be located closer to an interior front wall of a cookbox of the grill, and/or closer to a front lip of a flat top cooking surface of a griddle of the grill, as further described herein.

[0052] The above-identified features as well as other advantageous features of example space efficient gas valves for grills as disclosed herein are further described below in connection with the figures of the application.

[0053] As used herein, the term “configured” means sized, shaped, arranged, structured, oriented, positioned, and/or located. For example, in the context of a first part configured to fit within a second part, the first part is sized, shaped, arranged, structured, oriented, positioned, and/or located to fit within the second part.

[0054] As used herein in the context of a first object circumscribing a second object, the term “circumscribe” means that the first object is constructed around and/or defines an area around the second object. In interpreting the term “circumscribe” as used herein, it is to be understood that the first object circumscribing the second object can include gaps and/or can consist of multiple spaced-apart objects, such that a boundary formed by the first object around the second object is not necessarily a continuous boundary.

[0055] As used herein, unless otherwise stated, the terms “above” and “below” describe the relationship of two parts relative to Earth. For example, as used herein, a first part is “above” a second part if the second part is closer to Earth than the first part is. As another example, as used herein, a first part is “below” a second part if the first part is closer to Earth than the second part is. It is to be understood that a first part can be above or below a second part with one or more of: another part or parts therebetween; without another part therebetween; with the first and second

parts contacting one another; or without the first and second parts contacting one another.

[0056] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts at the point (or points) of contact between the two parts.

[0057] As used herein, the term “fastener” means any device(s), structure(s), and/or material(s) that is/are configured, individually or collectively, to couple, connect, attach, and/or fasten one or more component(s) to one or more other component(s). For example, a fastener can be implemented by any type(s) and/or any number(s) of bolts, nuts, screws, posts, anchors, rivets, pins, clips, ties, welds, adhesives, etc.

[0058] As used herein in the context of describing the relationship between two structures, the terms “in fluid communication,” “fluidically connected,” and/or “fluidically coupled” mean that the two structures are individually and/or collectively configured to allow a fluid (e.g., a gas or a liquid) to pass (e.g., to flow) from the first of the two structures to the second of the two structures, or vice-versa. For example, a second flow channel may be described as being in fluid communication with a first flow channel when a fluid (e.g., a gas or a liquid) is able to pass (e.g., to flow) from the first flow channel into the second flow channel, or from the second flow channel into the first flow channel.

[0059] As used herein, the terms “substantially” and/or “approximately” modify their subjects and/or values to recognize the potential presence of variations that occur in real world applications. For example, “substantially” and/or “approximately” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real-world imperfections as will be understood by persons of ordinary skill in the art. For example, “substantially” and/or “approximately” may indicate such dimensions may be within a tolerance range of $\pm 10\%$ unless otherwise specified in the description provided herein.

[0060] As used herein, the terms “including” and “comprising” (and all forms and tenses thereof) are open-ended terms. Thus, whenever the written description or a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation.

[0061] As used herein, singular references (e.g., “a,” “an,” “first,” “second,” etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements, or method actions may be implemented by, for example, the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0062] The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C.

[0063] As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open-ended. As used herein in the context of describing structures, components, items, objects, and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as

used herein in the context of describing structures, components, items, objects, and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0064] FIG. 1 is a first perspective view of an example gas valve **100** constructed in accordance with the teachings of this disclosure. FIG. 2 is a second perspective view of the gas valve **100** of FIG. 1. FIG. 3 is a right side view of the gas valve **100** of FIGS. 1 and 2. FIG. 4 is a left side view of the gas valve **100** of FIGS. 1-3. FIG. 5 is a front view of the gas valve **100** of FIGS. 1-4. FIG. 6 is a rear view of the gas valve **100** of FIGS. 1-5. FIG. 7 is a top view of the gas valve **100** of FIGS. 1-6. FIG. 8 is a bottom view of the gas valve **100** of FIGS. 1-7. The gas valve **100** of FIGS. 1-8 includes an example body **102**, a flow control member, an example stem **104**, and an example ignition assembly **106**. The body **102** of the gas valve **100** is configured to house, contain, carry, and/or support the flow control member, the stem **104**, and the ignition assembly **106** of the gas valve **100**, as further described herein.

[0065] In the illustrated example of FIGS. 1-8, the body **102** of the gas valve **100** includes an example inlet **202**, an example inlet conduit **204**, an example chamber housing **206**, an example first outlet conduit **108**, an example first outlet **208**, an example second outlet conduit **110**, and an example second outlet **210**. The inlet **202** of the body **102** is formed by and/or located at an example first end **212** of the inlet conduit **204**. The inlet **202** and/or the first end **212** of the inlet conduit **204** of the body **102** is/are configured to be coupled to a manifold such that a pressurized fluid (e.g., a pressurized gas) present in and/or flowing through the manifold is able to flow into the inlet **202** and/or into the first end **212** of the inlet conduit **204**. The inlet **202** of the body **102** is accordingly in fluid communication with the manifold, as further described herein.

[0066] The inlet conduit **204** of the body **102** of FIGS. 1-8 is coupled (e.g., at or proximate to an example second end **214** of the inlet conduit **204**) to the chamber housing **206** of the body **102**. The inlet conduit **204** is configured to transport and/or carry pressurized fluid (e.g., pressurized gas) from the inlet **202** of the body **102** toward and/or into the chamber housing **206** of the body **102**. In this regard, the inlet conduit **204** includes and/or defines an inlet flow channel that is fluidically coupled to and extends between the inlet **202** of the body **102** and a chamber of the chamber housing **206** of the body **102**. The inlet flow channel of the inlet conduit **204** is further described below in connection with FIGS. 9-11 and 20-25.

[0067] In the illustrated example of FIGS. 1-8, the inlet conduit **204** of the body **102** extends downwardly from the chamber housing **206** of the body **102** such that at least a portion (e.g., the first end **212**) of the inlet conduit **204** is located below the chamber housing **206**. In other examples, inlet conduit **204** of the body **102** can instead extend laterally from (e.g., to the right or to the left of) the chamber housing **206** of the body **102**. In the illustrated example of FIGS. 1-8, the inlet conduit **204** of the body **102** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the inlet conduit **204** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the inlet conduit **204** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. 1-8, the inlet conduit **204** of the body **102** includes a single linear segment. In other examples, the inlet conduit **204** can instead include a single curved or contoured segment. In still other examples, the inlet conduit **204** can instead include a plurality of segments of any types, sizes, and/or orientations, with respective ones of the segments being joined together to form the

inlet conduit **204** as a whole.

[0068] The chamber housing **206** of the body **102** of FIGS. **1-8** is coupled to the inlet conduit **204**, to first outlet conduit **108**, and to the second outlet conduit **110** of the body **102**. The chamber housing **206** is configured to transport and/or carry pressurized fluid (e.g., pressurized gas) from the inlet conduit **204** of the body **102** toward and/or into the first outlet conduit **108** of the body **102**, and/or toward and/or into the second outlet conduit **110** of the body **102**. In this regard, the chamber housing **206** includes and/or defines a chamber that is fluidically coupled to and extends between an inlet flow channel of the inlet conduit **204** of the body **102** on the one hand, and respective ones of a first outlet flow channel of the first outlet conduit **108** and a second outlet flow channel of the second outlet conduit **110** of the body **102** on the other hand. The chamber of the chamber housing **206** is configured to receive and/or contain the flow control member of the gas valve **100**, with the flow control member being rotatable relative to the chamber of the chamber housing **206** and/or, more generally, relative to the body **102** of the gas valve **100**. The chamber of the chamber housing **206** is further described below in connection with FIGS. **9-11** and **20-25**. An example flow control member that is configured to be received within the chamber of the chamber housing **206** is further described below in connection with FIGS. **12-25**.

[0069] In the illustrated example of FIGS. **1-8**, the chamber housing **206** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the chamber housing **206** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, a first portion (e.g., a front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the inlet conduit **204** of the body **102**, and a second portion (e.g., a rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the inlet conduit **204** of the body **102**. In other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the inlet conduit **204** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located rearward from and/or relative to, the inlet conduit **204** of the body **102**. In other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located forward from and/or relative to, the inlet conduit **204** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the inlet conduit **204** of the body **102**.

[0070] The first outlet conduit **108** of the body **102** of FIGS. **1-8** is coupled (e.g., at or proximate to an example first end **216** of the first outlet conduit **108**) to the chamber housing **206** of the body **102**. The first outlet conduit **108** is configured to transport and/or carry pressurized fluid (e.g., pressurized gas) from the chamber housing **206** of the body **102** toward and/or to the first outlet **208** of the body **102**. In this regard, the first outlet conduit **108** includes and/or defines a first outlet flow channel that is fluidically coupled to and extends between the chamber of the chamber housing **206** of the body **102** and the first outlet **208** of the body **102**. In some examples, a first portion of the first outlet flow channel is included within and/or defined by the chamber housing **206**, and a second portion of the first outlet flow channel downstream of the first portion of the first outlet flow channel is included within and/or defined by the first outlet conduit **108**. The first outlet flow channel of the first outlet conduit **108** is further described below in connection with FIGS. **9-11** and **20-25**.

[0071] In the illustrated example of FIGS. **1-8**, the first outlet conduit **108** of the body **102** includes an example first segment **112** and an example second segment **114**, with the second segment **114** being coupled to the first segment **112**. In other examples, the first outlet conduit **108** can include a different number (e.g., 1, 3, 4, etc.) of segments. In the illustrated example of FIGS. **1-8**, the first

segment **112** of the first outlet conduit **108** extends upwardly from the chamber housing **206** of the body **102** such that at least a portion of the first segment **112** of the first outlet conduit **108** is located above the chamber housing **206**. In other examples, the first segment **112** of the first outlet conduit **108** can instead extend laterally from (e.g., to the right or to the left of) the chamber housing **206** of the body **102**. In the illustrated example of FIGS. **1-8**, the first segment **112** of the first outlet conduit **108** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the first segment **112** of the first outlet conduit **108** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the first segment **112** of the first outlet conduit **108** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, the first segment **112** of the first outlet conduit **108** is linear. In other examples, the first segment **112** of the first outlet conduit **108** can instead be curved or contoured.

[0072] The second segment **114** of the first outlet conduit **108** of FIGS. **1-8** is configured to engage and/or to be coupled to an end portion of a burner tube. For example, as shown in FIGS. **1-8**, the second segment **114** of the first outlet conduit **108** is configured to be inserted into an open end of a burner tube such that the burner tube circumscribes the second segment **114** of the first outlet conduit **108**, as further described below. In the illustrated example of FIGS. **1-8**, the second segment **114** of the first outlet conduit **108** extends rearwardly from the first segment **112** of the first outlet conduit **108** such that at least a portion of the second segment **114** of the first outlet conduit **108** is located to the rear of the first segment **112** of the first outlet conduit **108**. In the illustrated example of FIGS. **1-8**, the second segment **114** of the first outlet conduit **108** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the second segment **114** of the first outlet conduit **108** can instead be oriented at an angle between a vertical orientation and a horizontal orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, the second segment **114** of the first outlet conduit **108** is linear. In other examples, the second segment **114** of the first outlet conduit **108** can instead be curved or contoured.

[0073] In the illustrated example of FIGS. **1-8**, a first portion (e.g., a front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the first segment **112** of the first outlet conduit **108** of the body **102**, and a second portion (e.g., a rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the first segment **112** of the first outlet conduit **108** of the body **102**. In other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the first segment **112** of the first outlet conduit **108** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located rearward from and/or relative to, the first segment **112** of the first outlet conduit **108** of the body **102**. In still other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located forward from and/or relative to, the first segment **112** of the first outlet conduit **108** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the first segment **112** of the first outlet conduit **108** of the body **102**.

[0074] The first outlet **208** of the body **102** is formed by and/or located at an example second end **218** of the first outlet conduit **108**. The first outlet **208** and/or the second end **218** of the first outlet conduit **108** of the body **102** is/are configured to be coupled to a burner tube such that a pressurized fluid (e.g., a pressurized gas) present in and/or flowing through the first outlet conduit **108** and/or the first outlet **208** is able to flow into the burner tube. The burner tube is accordingly in fluid communication with the first outlet **208** of the body **102**, as further described herein.

[0075] The second outlet conduit **110** of the body **102** of FIGS. **1-8** is coupled (e.g., at or proximate to an example first end **602** of the second outlet conduit **110**) to the chamber housing **206** of the body **102**. The second outlet conduit **110** is configured to transport and/or carry pressurized fluid (e.g., pressurized gas) from the chamber housing **206** of the body **102** toward and/or to the second outlet **210** of the body **102**. In this regard, the second outlet conduit **110** includes and/or defines a second outlet flow channel that is fluidically coupled to and extends between the chamber of the chamber housing **206** of the body **102** and the second outlet **210** of the body **102**. The second outlet flow channel of the second outlet conduit **110** is further described below in connection with FIGS. **9-11** and **20-25**.

[0076] In the illustrated example of FIGS. **1-8**, the second outlet conduit **110** of the body **102** includes an example first segment **220**, an example second segment **116** coupled to the first segment **220**, and an example third segment **118** coupled to the second segment **116**. In other examples, the second outlet conduit **110** can include a different number (e.g., 1, 2, 4, etc.) of segments. In the illustrated example of FIGS. **1-8**, the first segment **220** of the first outlet conduit **108** extends laterally from (e.g., to the right or to the left of) the chamber housing **206** of the body **102** such that at least a portion of the first segment **220** of the second outlet conduit **110** is located to the side of the chamber housing **206**. In other examples, the first segment **220** of the second outlet conduit **110** can instead extend upwardly from the chamber housing **206** of the body **102**. In the illustrated example of FIGS. **1-8**, the first segment **220** of the second outlet conduit **110** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the first segment **220** of the second outlet conduit **110** can instead be oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In still other examples, the first segment **220** of the second outlet conduit **110** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, the first segment **220** of the second outlet conduit **110** is linear. In other examples, the first segment **220** of the second outlet conduit **110** can instead be curved or contoured.

[0077] In the illustrated example of FIGS. **1-8**, the second segment **116** of the second outlet conduit **110** extends upwardly from the first segment **220** of the second outlet conduit **110** such that at least a portion of the second segment **116** of the second outlet conduit **110** is located above the first segment **220** of the second outlet conduit **110**. In other examples, the second segment **116** of the second outlet conduit **110** can instead extend laterally from (e.g., to the right or to the left of) the first segment **220** of the second outlet conduit **110**. In the illustrated example of FIGS. **1-8**, the second segment **116** of the second outlet conduit **110** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the second segment **116** of the second outlet conduit **110** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the second segment **116** of the second outlet conduit **110** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, the second segment **116** of the second outlet conduit **110** is linear. In other examples, the second segment **116** of the second outlet conduit **110** can instead be curved or contoured.

[0078] The third segment **118** of the second outlet conduit **110** of FIGS. **1-8** is configured to engage, be coupled to, and/or extend toward an end portion of an example ignition conduit **120** of the ignition assembly **106** of the gas valve **100**. For example, as shown in FIGS. **1-8**, the third segment **118** of the second outlet conduit **110** extends toward a front end of the ignition conduit **120**. In the illustrated example of FIGS. **1-8**, the third segment **118** of the second outlet conduit **110** extends rearwardly from the second segment **116** of the second outlet conduit **110** such that at least a portion of the third segment **118** of the second outlet conduit **110** is located to the rear of the second segment **116** of the second outlet conduit **110**. In the illustrated example of FIGS. **1-8**, the third segment **118** of the second outlet conduit **110** is oriented horizontally (e.g., parallel to a

horizontal plane). In other examples, the third segment **118** of the second outlet conduit **110** can instead be oriented at an angle between a vertical orientation and a horizontal orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, the third segment **118** of the second outlet conduit **110** is linear. In other examples, the third segment **118** of the second outlet conduit **110** can instead be curved or contoured.

[0079] In the illustrated example of FIGS. **1-8**, a first portion (e.g., a front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the first segment **220** of the second outlet conduit **110** of the body **102**, and a second portion (e.g., a rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the first segment **220** of the second outlet conduit **110** of the body **102**. In other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the first segment **220** of the second outlet conduit **110** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located rearward from and/or relative to, the first segment **220** of the second outlet conduit **110** of the body **102**. In still other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located forward from and/or relative to, the first segment **220** of the second outlet conduit **110** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the first segment **220** of the second outlet conduit **110** of the body **102**.

[0080] In the illustrated example of FIGS. **1-8**, a first portion (e.g., a front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the second segment **116** of the second outlet conduit **110** of the body **102**, and a second portion (e.g., a rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the second segment **116** of the second outlet conduit **110** of the body **102**. In other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located forward from and/or relative to the second segment **116** of the second outlet conduit **110** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located rearward from and/or relative to, the second segment **116** of the second outlet conduit **110** of the body **102**. In still other examples, the chamber housing **206** can instead be configured such the first portion (e.g., the front portion) of the chamber housing **206** of the body **102** does not extend or project, and/or is not located forward from and/or relative to, the second segment **116** of the second outlet conduit **110** of the body **102**, and the second portion (e.g., the rear portion) of the chamber housing **206** of the body **102** extends, projects, and/or is located rearward from and/or relative to the second segment **116** of the second outlet conduit **110** of the body **102**.

[0081] The second outlet **210** of the body **102** is formed by and/or located at an example second end **222** of the second outlet conduit **110**. The second outlet **210** and/or the second end **222** of the second outlet conduit **110** of the body **102** is/are configured to be coupled to the ignition conduit **120** of the ignition assembly **106** of the gas valve **100** such that a pressurized fluid (e.g., a pressurized gas) present in and/or flowing through the second outlet conduit **110** and/or the second outlet **210** is able to flow into the ignition conduit **120**. The ignition conduit **120** is accordingly in fluid communication with the second outlet **210** of the body **102**, as further described herein.

[0082] The stem **104** of the gas valve **100** of FIGS. **1-8** is rotatable relative to the body **102** of the gas valve **100**. In the illustrated example of FIGS. **1-8** the stem **104** extends forwardly from the body **102** such that at least a portion (e.g., an example front end **122**) of the stem **104** is located in

front of the body **102**. A first portion (e.g., the front end **122**) of the stem **104** is configured to be operatively and/or mechanically coupled to a control knob of a grill such that a rotation (e.g., a clockwise rotation or a counter-clockwise rotation) of the control knob about an axis of rotation of the control knob causes a corresponding rotation of the stem **104** about an axis of rotation of the stem **104**. In some examples, the axis of rotation of the stem and the axis of rotation of the control knob are coaxially arranged relative to one another. A second portion (e.g., a rear end) of the stem **104** is configured to be operatively and/or mechanically coupled to the flow control member of the gas valve **100** such that a rotation (e.g., a clockwise rotation or a counter-clockwise rotation) of the stem **104** about an axis of rotation of the stem **104** causes a corresponding rotation of the flow control member about an axis of rotation of the flow control member, whereby rotation of the flow control member occurs within and/or relative to the chamber of the chamber housing **206** of the body **102** of the gas valve **100**. In some examples, the axis of rotation of the stem and the axis of rotation of the flow control member are coaxially arranged relative to one another. In the illustrated example of FIGS. **1-8**, the stem **104** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the stem **104** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0083] The ignition assembly **106** of the gas valve **100** of FIGS. **1-8** is configured to ignite and/or induce the ignition of combustible fluid (e.g., combustible gas) that passes and/or flows through the body **102** of the gas valve **100**. In the illustrated example of FIGS. **1-8**, the ignition assembly **106** includes the ignition conduit **120** described above, and further includes an example spark generator **124**. The ignition conduit **120** is coupled (e.g., via a clip) to the third segment **118** of the second outlet conduit **110** of the body **102** of the gas valve **100**. In the illustrated example of FIGS. **1-8**, the ignition conduit **120** extends rearwardly from the third segment **118** of the second outlet conduit **110** such that at least a portion of the ignition conduit **120** is located to the rear of the third segment **118** of the second outlet conduit **110**. In the illustrated example of FIGS. **1-8**, the ignition conduit **120** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the ignition conduit **120** can instead be oriented at an angle between a vertical orientation and a horizontal orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **1-8**, the ignition conduit **120** is linear. In other examples, the ignition conduit **120** can instead be curved or contoured.

[0084] In the illustrated example of FIGS. **1-8**, the ignition conduit **120** includes an inlet located at an example first end **126** of the ignition conduit **120**, and further includes an example outlet **128** located at an example second end **130** of the ignition conduit **120**. The inlet of the ignition conduit **120** is in fluid communication with the second outlet **210** of the body **102** of the gas valve **100** such that pressurized fluid (e.g., pressurized gas) exiting the body **102** of the gas valve **100** via the second outlet **210** of the body **102** passes (e.g., flows) into the inlet of the ignition conduit **120**. The spark generator **124** of the ignition assembly **106** is coupled (e.g., via a clip) to the ignition conduit **120**, with the spark generator **124** being oriented and/or positioned relative to the ignition conduit **120** such that an example tip **132** of the spark generator **124** is located adjacent and/or proximate to the outlet **128** of the ignition conduit **120**, thereby enabling the spark generator **124** to ignite combustible gas (e.g., via a spark provided by the tip **132** of the spark generator **124** as the combustible gas exits the outlet **128** of the ignition conduit **120**).

[0085] FIG. **9** is a partial cutaway view of the gas valve **100** of FIGS. **1-8** taken along section A-A of FIG. **5**, with the flow control member of the gas valve **100** omitted for enhanced viewability. FIG. **10** is a cross-sectional view of the gas valve **100** of FIGS. **1-9** taken along section A-A of FIG. **5**, with the flow control member of the gas valve **100** omitted for enhanced viewability. FIG. **11** is a partial cutaway view of the gas valve **100** of FIGS. **1-10** taken along section B-B of FIG. **7**, with the flow control member of the gas valve **100** omitted for enhanced viewability. As shown in FIGS. **9-11**, the body **102** of the gas valve **100** includes and/or defines an example gas train **900** extending

from the inlet **202** of the body **102** to respective ones of the first outlet **208** and the second outlet **210** of the body **102**. In the illustrated example of FIGS. **9-11**, the gas train **900** includes the inlet **202**, an example inlet flow channel **902**, an example chamber **904**, an example first outlet flow channel **906**, the first outlet **208**, an example second outlet flow channel **1102**, and the second outlet **210**.

[0086] The inlet **202** of the gas train **900** of FIGS. **9-11** is formed by, defined by, and/or located at the first end **212** of the inlet conduit **204** of the body **102**. The inlet **202** of the gas train **900** is configured to be in fluid communication with an outlet and/or an opening of a manifold such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the manifold is able to flow from the manifold into and/or through the inlet **202**. The inlet flow channel **902** of the gas train **900** of FIGS. **9-11** is formed by, defined by, and/or located within the inlet conduit **204** of the body **102**. The inlet flow channel **902** of the gas train **900** is in fluid communication with the inlet **202** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the inlet **202** is able to flow from the inlet **202** into the inlet flow channel **902**.

[0087] As shown in FIGS. **9-11**, the inlet flow channel **902** of the gas train **900** extends downwardly from the chamber **904** of the gas train **900** such that at least a portion of the inlet flow channel **902** is located below the chamber **904**. In other examples, inlet flow channel **902** of the gas train **900** can instead extend laterally from (e.g., to the right or to the left of) the chamber **904** of the gas train **900**. In the illustrated example of FIGS. **9-11**, the inlet flow channel **902** of the gas train **900** includes an example linear segment **908** having an example central axis **1002**. As shown in FIGS. **9-11**, the central axis **1002** of the linear segment **908** of the inlet flow channel **902** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the central axis **1002** of the linear segment **908** of the inlet flow channel **902** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the central axis **1002** of the linear segment **908** of the inlet flow channel **902** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0088] The chamber **904** of the gas train **900** of FIGS. **9-11** is formed by, defined by, and/or located within the chamber housing **206** of the body **102**. In the illustrated example of FIGS. **9-11**, the chamber **904** includes an example inlet opening **910**, an example first outlet opening **912**, and an example second outlet opening **914**. The chamber **904** of the gas train **900** is in fluid communication with the inlet flow channel **902** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the inlet flow channel **902** is able to flow from the inlet flow channel **902** into the chamber **904** via the inlet opening **910** of the chamber **904**. The entry, passage, and/or flow of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** into the chamber **904** via the inlet opening **910** of the chamber **904** occurs selectively based on the rotational position of the flow control member of the gas valve **100** within the chamber **904**, as further described herein.

[0089] As shown in FIGS. **9-11**, the first outlet opening **912** and the second outlet opening **914** of the chamber **904** respectively lead to separate exit pathways defined by the gas train **900**. The first exit pathway of the gas train **900** of FIGS. **9-11**, which originates at the first outlet opening **912** of the chamber **904**, includes the first outlet flow channel **906** and the first outlet **208**. The chamber **904** of the gas train **900** is in fluid communication with the first outlet flow channel **906** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the chamber **904** is able to flow from the chamber **904** into the first outlet flow channel **906** via the first outlet opening **912** of the chamber **904**. The entry, passage, and/or flow of pressurized fluid (e.g., pressurized gas) from the chamber **904** into the first outlet flow channel **906** via the first outlet opening **912** of the chamber **904** occurs selectively based on the rotational position of the flow control member of the gas valve **100** within the chamber **904**, as further described herein.

[0090] The second exit pathway of the gas train **900** of FIGS. **9-11**, which originates at the second

outlet opening **914** of the chamber **904**, includes the second outlet flow channel **1102** and the second outlet **210**. The chamber **904** of the gas train **900** is in fluid communication with the second outlet flow channel **1102** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the chamber **904** is able to flow from the chamber **904** into the second outlet flow channel **1102** via the second outlet opening **914** of the chamber **904**. The entry, passage, and/or flow of pressurized fluid (e.g., pressurized gas) from the chamber **904** into the second outlet flow channel **1102** via the second outlet opening **914** of the chamber **904** occurs selectively based on the rotational position of the flow control member of the gas valve **100** within the chamber **904**, as further described herein.

[0091] As shown in FIGS. **9-11**, the chamber **904** of the gas train **900** is configured to receive and/or contain the flow control member of the gas valve **100**, with the flow control member being rotatable within and/or relative to the chamber **904** and/or, more generally, relative to the body **102** of the gas valve **100**. In the illustrated example of FIGS. **9-11**, the chamber **904** has a conical chape configured to receive and/or contain a conically-shaped flow control member (e.g., a cone). An example conically-shaped flow control member that is configured to be received within the chamber **904** is further described below in connection with FIGS. **12-25**. In other examples, the chamber **904** can instead have a cylindrical shape configured to receive a cylindrically-shaped flow control member, or a spherical shape configured to receive a spherically-shaped flow control member.

[0092] The chamber **904** of the gas train **900** of FIGS. **9-11** has an example central axis **1004**. In the illustrated example of FIGS. **9-11**, the central axis **1004** of the chamber **904** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the central axis **1004** of the chamber **904** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane). In the illustrated example of FIGS. **9-11**, the central axis **1004** of the chamber **904** is parallel to and is coaxially aligned with an example axis of rotation **1006** of the stem **104** of the gas valve **100**. In other examples, the axis of rotation **1006** of the stem **104** of the gas valve **100** can instead be offset from and/or angled relative to the central axis **1004** of the chamber **904**. In the illustrated example of FIGS. **9-11**, the inlet opening **910** of the chamber **904** is located between the first outlet opening **912** of the chamber **904** and the second outlet opening **914** of the chamber **904**. In other examples, the first outlet opening **912** of the chamber **904** can be located between the inlet opening **910** of the chamber **904** and the second outlet opening **914** of the chamber **904**. In still other examples, the second outlet opening **914** of the chamber **904** can be located between the inlet opening **910** of the chamber **904** and the first outlet opening **912** of the chamber **904**.

[0093] The first outlet flow channel **906** of the gas train **900** of FIGS. **9-11** is formed by, defined by, and/or located within the chamber housing **206** and/or the first outlet conduit **108** of the body **102**. For example, as shown in FIGS. **9-11**, a first portion of the first outlet flow channel **906** of the gas train **900** is formed by, defined by, and/or located within the chamber housing **206** of the body **102**, and a second portion of the first outlet flow channel **906** of the gas train **900** located downstream from the first portion of the first outlet flow channel **906** of the gas train **900** is formed by, defined by, and/or located within the first outlet conduit **108**. The first outlet flow channel **906** of the gas train **900** is in fluid communication with the chamber **904** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the chamber **904** is able to flow from the chamber **904** into the first outlet flow channel **906** via the first outlet opening **912** of the chamber **904**. The entry, passage, and/or flow of pressurized fluid (e.g., pressurized gas) from the chamber **904** into the first outlet flow channel **906** via the first outlet opening **912** of the chamber **904** occurs selectively based on the rotational position of the flow control member of the gas valve **100** within the chamber **904**, as further described herein.

[0094] As shown in FIGS. **9-11**, the first outlet flow channel **906** of the gas train **900** extends upwardly from the chamber **904** of the gas train **900** such that at least a portion of the first outlet

flow channel **906** is located above the chamber **904**. In other examples, the first outlet flow channel **906** of the gas train **900** can instead extend laterally from (e.g., to the right or to the left of) the chamber **904** of the gas train **900**. In the illustrated example of FIGS. **9-11**, the first outlet flow channel **906** of the gas train **900** includes an example first linear segment **916**, an example second linear segment **918**, an example third linear segment **920**, and an example fourth linear segment **922**. The first linear segment **916** of the first outlet flow channel **906** is in fluid communication with and located downstream of the chamber **904**. The second linear segment **918** of the first outlet flow channel **906** is in fluid communication with and located downstream of the first linear segment **916** of the first outlet flow channel **906**. The third linear segment **920** of the first outlet flow channel **906** is in fluid communication with and located downstream of the second linear segment **918** of the first outlet flow channel **906**. The fourth linear segment **922** of the first outlet flow channel **906** is in fluid communication with and located downstream of the third linear segment **920** of the first outlet flow channel **906**.

[0095] In the illustrated example of FIGS. **9-11**, the first linear segment **916** of the first outlet flow channel **906** extends upwardly from the chamber **904** of the gas train **900** such that at least a portion of the first linear segment **916** of the first outlet flow channel **906** is located above the chamber **904**. The first linear segment **916** of the first outlet flow channel **906** has an example central axis **1008**. As shown in FIGS. **9-11**, the central axis **1008** of the first linear segment **916** of the first outlet flow channel **906** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the central axis **1008** of the first linear segment **916** of the first outlet flow channel **906** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the central axis **1008** of the first linear segment **916** of the first outlet flow channel **906** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0096] In the illustrated example of FIGS. **9-11**, the second linear segment **918** of the first outlet flow channel **906** extends forwardly from the first linear segment **916** of the first outlet flow channel **906** such that at least a portion of the second linear segment **918** of the first outlet flow channel **906** is located in front of the first linear segment **916** of the first outlet flow channel **906**. The second linear segment **918** of the first outlet flow channel **906** has an example central axis **1010**. As shown in FIGS. **9-11**, the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906** can instead be oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In still other examples, the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0097] In the illustrated example of FIGS. **9-11**, the third linear segment **920** of the first outlet flow channel **906** extends upwardly from the second linear segment **918** of the first outlet flow channel **906** such that at least a portion of the third linear segment **920** of the first outlet flow channel **906** is located above the second linear segment **918** of the first outlet flow channel **906**. The third linear segment **920** of the first outlet flow channel **906** has an example central axis **1012**. As shown in FIGS. **9-11**, the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0098] In the illustrated example of FIGS. **9-11**, the fourth linear segment **922** of the first outlet

flow channel **906** extends rearwardly from the third linear segment **920** of the first outlet flow channel **906** such that at least a portion of the fourth linear segment **922** of the first outlet flow channel **906** is located to the rear of the third linear segment **920** of the first outlet flow channel **906**. The fourth linear segment **922** of the first outlet flow channel **906** has an example central axis **1014**. As shown in FIGS. **9-11**, the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0099] The first outlet **208** of the gas train **900** of FIGS. **9-11** is formed by, defined by, and/or located at the second end **218** of the first outlet conduit **108** of the body **102**. The first outlet **208** of the gas train **900** is in fluid communication with the first outlet flow channel **906** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the first outlet flow channel **906** (e.g., flowing sequentially through the first linear segment **916**, the second linear segment **918**, the third linear segment **920**, and the fourth linear segment **922** of the first outlet flow channel **906**) is able to flow from the first outlet flow channel **906** into and/or through the first outlet **208**. The first outlet **208** of the gas train **900** is also configured to be in fluid communication with an open end of a burner tube operatively positioned downstream from the first outlet **208** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the first outlet **208** is able to flow from the first outlet **208** into the burner tube.

[0100] In the illustrated example of FIGS. **9-11**, the central axis of the **1004** of the chamber **904**, the central axis **1008** of the first linear segment **916** of the first outlet flow channel **906**, the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906**, the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906**, and the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** are coplanar. The central axis of the **1004** of the chamber **904**, the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906**, and the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** are horizontally oriented and parallel to one another, with the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906** being offset from and located above the central axis of the **1004** of the chamber **904**, and with the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** being offset from and located above the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906**. The central axis **1008** of the first linear segment **916** of the first outlet flow channel **906** and the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906** are vertically oriented and parallel to one another, with the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906** being located forward of the central axis **1008** of the first linear segment **916** of the first outlet flow channel **906**. Furthermore, respective ones of the central axis of the **1004** of the chamber **904**, the central axis **1010** of the second linear segment **918** of the first outlet flow channel **906**, and the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** are oriented orthogonally (e.g., perpendicularly) relative to respective ones of the central axis **1008** of the first linear segment **916** of the first outlet flow channel **906** and the central axis **1012** of the third linear segment **920** of the first outlet flow channel **906**.

[0101] The above-described arrangement of parts results in the chamber **904**, the first linear segment **916** of the first outlet flow channel **906**, the second linear segment **918** of the first outlet flow channel **906**, the third linear segment **920** of the first outlet flow channel **906**, and the fourth linear segment **922** of the first outlet flow channel **906** forming, defining, and/or otherwise providing a serpentine (e.g., S-shaped) portion of the gas train **900**. The serpentine (e.g., S-shaped) portion of the gas train **900** of FIGS. **9-11** advantageously reduces (e.g., minimizes) the front-to-rear distance between the inlet **202** of the gas valve **100** and the first outlet **208** of the gas valve **100**, and/or the front-to-rear distance between the stem **104** of the gas valve **100** and the first outlet

208 of the gas valve **100**. The improved space efficiency associated with this/these reduced dimension(s) of the gas valve **100** advantageously enables the gas valve **100** to be disposed within a correspondingly smaller (e.g., reduced depth) control panel chamber of a grill, and also advantageously enables a burner tube of such a grill to be located closer to an interior front wall of a cookbox of such a grill, as further described herein.

[0102] The second outlet flow channel **1102** of the gas train **900** of FIGS. **9-11** is formed by, defined by, and/or located within the second outlet conduit **110** of the body **102**. The second outlet flow channel **1102** of the gas train **900** is in fluid communication with the chamber **904** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the chamber **904** is able to flow from the chamber **904** into the second outlet flow channel **1102** via the second outlet opening **914** of the chamber **904**. The entry, passage, and/or flow of pressurized fluid (e.g., pressurized gas) from the chamber **904** into the second outlet flow channel **1102** via the second outlet opening **914** of the chamber **904** occurs selectively based on the rotational position of the flow control member of the gas valve **100** within the chamber **904**, as further described herein.

[0103] As shown in FIGS. **9-11**, the second outlet flow channel **1102** of the gas train **900** extends both laterally and upwardly from the chamber **904** of the gas train **900**. For example, as shown in FIGS. **9-11**, a first portion of the second outlet flow channel **1102** is located to the side of the chamber **904**, and a second portion of the second outlet flow channel **1102** located downstream of the first portion of the second outlet flow channel **1102** is located above the chamber **904**. In other examples, the first portion of the second outlet flow channel **1102** can instead extend upwardly from the chamber **904**, and the second portion of the second outlet flow channel **1102** can instead extend to the side of the chamber **904**. In the illustrated example of FIGS. **9-11**, the second outlet flow channel **1102** of the gas train **900** includes an example first linear segment **1104**, an example second linear segment **1106**, and an example third linear segment **1108**. The first linear segment **1104** of the second outlet flow channel **1102** is in fluid communication with and located downstream of the chamber **904**. The second linear segment **1106** of the second outlet flow channel **1102** is in fluid communication with and located downstream of the first linear segment **1104** of the second outlet flow channel **1102**. The third linear segment **1108** of the second outlet flow channel **1102** is in fluid communication with and located downstream of the second linear segment **1106** of the second outlet flow channel **1102**.

[0104] In the illustrated example of FIGS. **9-11**, the first linear segment **1104** of the second outlet flow channel **1102** extends laterally from (e.g., to the right or to the left of) the chamber **904** of the gas train **900** such that at least a portion of the first linear segment **1104** of the second outlet flow channel **1102** is located to the side of the chamber **904**. As shown in FIGS. **9-11**, the first linear segment **1104** of the second outlet flow channel **1102** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the first linear segment **1104** of the second outlet flow channel **1102** can instead be oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In still other examples, the first linear segment **1104** of the second outlet flow channel **1102** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0105] In the illustrated example of FIGS. **9-11**, the second linear segment **1106** of the second outlet flow channel **1102** extends upwardly from the first linear segment **1104** of the second outlet flow channel **1102** such that at least a portion of the second linear segment **1106** of the second outlet flow channel **1102** is located above the first linear segment **1104** of the second outlet flow channel **1102**. As shown in FIGS. **9-11**, the second linear segment **1106** of the second outlet flow channel **1102** is oriented vertically (e.g., at ninety degrees relative to a horizontal plane). In other examples, the second linear segment **1106** of the second outlet flow channel **1102** can instead be oriented horizontally (e.g., parallel to a horizontal plane). In still other examples, the second linear segment **1106** of the second outlet flow channel **1102** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty

degrees, etc. relative to a horizontal plane).

[0106] In the illustrated example of FIGS. 9-11, the third linear segment **1108** of the second outlet flow channel **1102** extends rearwardly from the second linear segment **1106** of the second outlet flow channel **1102** such that at least a portion of the third linear segment **1108** of the second outlet flow channel **1102** is located to the rear of the second linear segment **1106** of the second outlet flow channel **1102**. As shown in FIGS. 9-11, the third linear segment **1108** of the second outlet flow channel **1102** is oriented horizontally (e.g., parallel to a horizontal plane). In other examples, the third linear segment **1108** of the second outlet flow channel **1102** can instead be oriented at an angle between a horizontal orientation and a vertical orientation (e.g., at thirty degrees, forty-five degrees, sixty degrees, etc. relative to a horizontal plane).

[0107] The second outlet **210** of the gas train **900** of FIGS. 9-11 is formed by, defined by, and/or located at the second end **222** of the second outlet conduit **110** of the body **102**. The second outlet **210** of the gas train **900** is in fluid communication with the second outlet flow channel **1102** of the gas train **900** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the second outlet flow channel **1102** (e.g., flowing sequentially through the first linear segment **1104**, the second linear segment **1106**, and the third linear segment **1108** of the second outlet flow channel **1102**) is able to flow from the second outlet flow channel **1102** into and/or through the second outlet **210**. The second outlet **210** of the gas train **900** is also configured to be in fluid communication with the first end **126** of the ignition conduit **120** of the gas valve **100** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the second outlet **210** is able to flow from the second outlet **210** into the ignition conduit **120**.

[0108] FIG. 12 is a first perspective view of an example cone **1200** constructed in accordance with the teachings of this disclosure. FIG. 13 is a second perspective view of the cone **1200** of FIG. 12. FIG. 14 is a right side view of the cone **1200** of FIGS. 12 and 13. FIG. 15 is a left side view of the cone **1200** of FIGS. 12-14. FIG. 16 is a front view of the cone **1200** of FIGS. 12-15. FIG. 17 is a rear view of the cone **1200** of FIGS. 12-16. FIG. 18 is a top view of the cone **1200** of FIGS. 12-17. FIG. 19 is a bottom view of the cone **1200** of FIGS. 12-18. The cone **1200** of FIGS. 12-19 can be implemented as the flow control member of the gas valve **100** of FIGS. 1-11 described above. In this regard, the cone **1200** of FIGS. 12-19 includes an example sidewall **1202** having an example outer surface **1204** and an example inner surface **1302**. The outer surface **1204** of the sidewall **1202** of the cone **1200** has a conical shape that is configured to engage and/or fit snugly within, but remains rotatable relative to, the conically-shaped chamber **904** of the gas train **900** of the gas valve **100**. The inner surface **1302** of the sidewall **1202** of the cone **1200** is located radially inward of the outer surface **1204** of the sidewall **1202** of the cone **1200**. The inner surface **1302** of the sidewall **1202** of the cone **1200** defines an example flow chamber **1304** of the cone **1200**.

[0109] The cone **1200** of FIGS. 12-19 further includes an example first end **1206** (e.g., a front end) and an example second end **1208** (e.g., a rear end) located opposite the first end **1206**. The first end **1206** of the cone **1200** is closed such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the flow chamber **1304** of the cone **1200** is not able to flow out of the flow chamber **1304** through the first end **1206** of the cone **1200**. The second end **1208** of the cone **1200** is open such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the flow chamber **1304** of the cone **1200** is able to flow out of the flow chamber **1304** through the second end **1208** of the cone **1200**.

[0110] The cone **1200** of FIGS. 12-19 has an example central axis **1402** extending between the first end **1206** and the second end **1208** of the cone **1200**. The central axis **1402** of the cone **1200** also serves as an axis of rotation of the cone **1200**. In this regard, the central axis **1402** of the cone **1200** is coaxially positioned relative to the central axis **1004** of the chamber **904** of the gas train **900** of the gas valve **100** when the cone **1200** is disposed within the chamber **904**. When so disposed, the central axis **1402** of the cone **1200** is also coaxially positioned relative to the axis of rotation **1006** of the stem **104** of the gas valve **100**. In the illustrated example of FIGS. 12-19, the first end **1206**

of the cone **1200** includes example flanges **1210** configured to be mechanically engaged by and/or mechanically coupled to one or more intervening structural component(s) (e.g., a link, a rod, a pin, a fitting, a spring, etc.) that is/are in turn mechanically coupled to the stem **104** of the gas valve **100** such that rotational movement of the stem **104** about the axis of rotation **1006** of the stem **104** causes a corresponding rotational movement of the cone **1200** about the central axis **1402** of the cone **1200**, and/or about the central axis **1004** of the chamber **904** of the gas valve **100** when the cone **1200** is disposed in the chamber **904**. In other examples, the stem **104** of the gas valve **100** can instead be directly mechanically coupled to the first end **1206** of the cone **1200** such that rotational movement of the stem **104** about the axis of rotation **1006** of the stem **104** causes a corresponding rotational movement of the cone **1200** about the central axis **1402** of the cone **1200**, and/or about the central axis **1004** of the chamber **904** of the gas valve **100** when the cone **1200** is disposed in the chamber **904**.

[0111] The cone **1200** of FIGS. **12-19** further includes an example first inlet opening **1502**, an example second inlet opening **1212**, and an example third inlet opening **1214**, each of which extends through the sidewall **1202** of the cone **1200** (e.g., from the outer surface **1204** of the sidewall **1202** to the inner surface **1302** of the sidewall **1202**). In the illustrated example of FIGS. **12-19**, the diameter of the first inlet opening **1502** of the cone **1200** is greater than the diameter of the second inlet opening **1212** of the cone **1200**, and the diameter of the second inlet opening **1212** of the cone **1200** is greater than the diameter of the third inlet opening **1214** of the cone **1200**. In other examples, the diameter of the first inlet opening **1502** of the cone **1200** can instead be less than the diameter of the second inlet opening **1212** of the cone **1200**, and the diameter of the second inlet opening **1212** of the cone **1200** can instead be less than the diameter of the third inlet opening **1214** of the cone **1200**. In the illustrated example of FIGS. **12-19**, the first inlet opening **1502**, the second inlet opening **1212**, and the third inlet opening **1214** of the cone **1200** are spaced apart from one another and circumferentially arranged about the sidewall **1202** of the cone **1200**. In other examples, the first inlet opening **1502**, the second inlet opening **1212**, and the third inlet opening **1214** of the cone **1200** can instead be fluidically connected to one another (e.g., via a slot or channel) to create a continuous (e.g., slotted) inlet opening that extends circumferentially about an arc portion of the sidewall **1202** of the cone **1200**.

[0112] The cone **1200** of FIGS. **12-19** further includes an example first outlet opening **1306** and an example second outlet opening **1216**. The first outlet opening **1306** of the cone **1200** is formed and/or located at the second end **1208** (e.g., the rear end) of the cone **1200**. The second outlet opening **1216** of the cone **1200** extends through the sidewall **1202** of the cone **1200** (e.g., from the outer surface **1204** of the sidewall **1202** to the inner surface **1302** of the sidewall **1202**). In the illustrated example of FIGS. **12-19**, the diameter of the first outlet opening **1306** of the cone **1200** is greater than the diameter of the second outlet opening **1216** of the cone **1200**. In other examples, the diameter of the first outlet opening **1306** of the cone **1200** can instead be less than the diameter of the second outlet opening **1216** of the cone **1200**. In the illustrated example of FIGS. **12-19**, the second outlet opening **1216** of the cone **1200** is located forward of each of the first inlet opening **1502**, the second inlet opening **1212**, and the third inlet opening **1214** of the cone **1200**. In other examples, the second outlet opening **1216** of the cone **1200** can instead be located rearward of the first inlet opening **1502**, the second inlet opening **1212**, and/or the third inlet opening **1214** of the cone **1200**.

[0113] When the cone **1200** of FIGS. **12-19** is disposed within the chamber **904** of the gas train **900**, movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** of the gas train **900** into the chamber **904** of the gas train **900** and/or into the flow chamber **1304** of the cone **1200**, and movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** and/or the chamber **904** of the gas train **900** into respective ones of the first outlet flow channel **906** and/or the second outlet flow channel **1102** of the gas train **900** is controlled by the rotational position of the cone **1200** relative to the chamber **904**. In this regard, the cone **1200** of

the gas valve **100** is rotatable relative to the chamber **904** of the gas valve **100** between a closed position and one or more open position(s). Placing the cone **1200** in the closed position prevents the movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** into the chamber **904** and/or the flow chamber **1304**, and/or prevents the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** and/or the chamber **904** into respective ones of the first outlet flow channel **906** and/or the second outlet flow channel **1102**. Conversely, placing the cone **1200** in one of the open position(s) enables the movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** into the chamber **904** and/or the flow chamber **1304**, and/or enables the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** and/or the chamber **904** into respective ones of the first outlet flow channel **906** and/or the second outlet flow channel **1102**.

[0114] FIG. **20** is a partial cutaway view taken along section A-A of FIG. **5**, showing the cone **1200** of FIGS. **12-19** disposed in the chamber **904** of the gas valve **100** of FIGS. **1-11**, with the cone **1200** in an example closed position **2000**. FIG. **21** is a cross-sectional view taken along section A-A of FIG. **5**, showing the cone **1200** of FIGS. **12-20** disposed in the chamber **904** of the gas valve **100** of FIGS. **1-11** and **20**, with the cone **1200** in the closed position **2000** of FIG. **20**. FIG. **22** is a partial cutaway view taken along section B-B of FIG. **7**, showing the cone **1200** of FIGS. **12-21** disposed in the chamber **904** of the gas valve **100** of FIGS. **1-11**, **20**, and **21**, with the cone **1200** in the closed position **2000** of FIGS. **20** and **21**.

[0115] When the cone **1200** is positioned in the closed position **2000** shown in FIGS. **20-22** relative to the chamber **904** of the gas train **900**, none of the inlet openings (e.g., the first inlet opening **1502**, the second inlet opening **1212**, and the third inlet opening **1214**) formed in the sidewall **1202** of the cone **1200** is aligned with the inlet opening **910** of the chamber **904**. The inlet opening **910** of the chamber **904** is accordingly blocked and/or covered by a solid portion of the sidewall **1202** of the cone **1200**, thereby preventing the flow chamber **1304** of the cone **1200** from being in fluid communication with the inlet flow channel **902** of the gas train **900**, and thereby also preventing the movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** of the gas train **900** into the flow chamber **1304** of the cone **1200** via the inlet opening **910** of the chamber **904**.

[0116] When the cone **1200** is positioned in the closed position **2000** shown in FIGS. **20-22** relative to the chamber **904** of the gas train **900**, there is also a lack of alignment between the second outlet opening **1216** formed in the sidewall **1202** of the cone **1200** and the second outlet opening **914** of the chamber **904**. The second outlet opening **914** of the chamber **904** is accordingly blocked and/or covered by a solid portion of the sidewall **1202** of the cone **1200**, thereby preventing the flow chamber **1304** of the cone **1200** from being in fluid communication with the second outlet flow channel **1102** of the gas train **900**, and thereby also preventing the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the second outlet flow channel **1102** of the gas train **900** via the second outlet opening **914** of the chamber **904**.

[0117] FIG. **23** is a partial cutaway view taken along section A-A of FIG. **5**, showing the cone **1200** of FIGS. **12-22** disposed in the chamber **904** of the gas valve **100** of FIGS. **1-11** and **20-22**, with the cone **1200** in an example open position **2300**. FIG. **24** is a cross-sectional view taken along section A-A of FIG. **5**, showing the cone **1200** of FIGS. **12-23** disposed in the chamber **904** of the gas valve **100** of FIGS. **1-11** and **20-23**, with the cone **1200** in the open position **2300** of FIG. **23**. FIG. **25** is a partial cutaway view taken along section B-B of FIG. **7**, showing the cone **1200** of FIGS. **12-24** disposed in the chamber **904** of the gas valve **100** of FIGS. **1-11** and **20-24**, with the cone **1200** in the open position **2300** of FIGS. **23** and **24**.

[0118] When the cone **1200** is positioned in the open position **2300** shown in FIGS. **23-25** relative to the chamber **904** of the gas train **900**, the first inlet opening **1502** formed in the sidewall **1202** of the cone **1200** is aligned with the inlet opening **910** of the chamber **904**. The flow chamber **1304** of the cone **1200** is accordingly in fluid communication with the inlet flow channel **902** of the gas

train **900** (e.g., via the alignment between the first inlet opening **1502** of the cone **1200** and the inlet opening **910** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** of the gas train **900** into the flow chamber **1304** of the cone **1200** via the inlet opening **910** of the chamber **904** and the first inlet opening **1502** of the cone **1200**. When the cone **1200** is positioned in the open position **2300** shown in FIGS. 23-25 relative to the chamber **904** of the gas train **900**, the flow chamber **1304** of the cone **1200** is also in fluid communication with the first outlet flow channel **906** of the gas train **900** (e.g., via the first outlet opening **1306** of the cone **1200** and the first outlet opening **912** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the first outlet flow channel **906** of the gas train **900** via the first outlet opening **1306** of the cone **1200** and the first outlet opening **912** of the chamber **904**.

[0119] When the cone **1200** is positioned in the open position **2300** shown in FIGS. 23-25 relative to the chamber **904** of the gas train **900**, the second outlet opening **1216** formed in the sidewall **1202** of the cone **1200** is aligned with the second outlet opening **914** of the chamber **904**. The flow chamber **1304** of the cone **1200** is accordingly in fluid communication with the second outlet flow channel **1102** of the gas train **900** (e.g., via the alignment between the second outlet opening **1216** of the cone **1200** and the second outlet opening **914** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the second outlet flow channel **1102** of the gas train **900** via the second outlet opening **1216** of the cone **1200** and the second outlet opening **914** of the chamber **904**. Positioning the cone **1200** in the open position **2300** shown in FIGS. 23-25 can accordingly promote and/or cause the ignition of pressurized combustible fluid (e.g., pressurized combustible gas) that flows through the second outlet flow channel **1102** of the gas train **900** into the ignition conduit **120** of the ignition assembly **106** of the gas valve **100**, as described above.

[0120] The cone **1200** can be rotated relative to the chamber **904** of the gas train **900** to other open positions that differ from the open position **2300** of FIGS. 23-25 described above. For example, the cone **1200** can be rotated relative to the chamber **904** of the gas train **900** by an additional ninety degree counterclockwise relative to the open position **2300** shown in FIGS. 23-25, thereby placing the cone **1200** in a second open position. When the cone **1200** is positioned in such a second open position, the second inlet opening **1212** formed in the sidewall **1202** of the cone **1200** is aligned with the inlet opening **910** of the chamber **904**. The flow chamber **1304** of the cone **1200** is accordingly in fluid communication with the inlet flow channel **902** of the gas train **900** (e.g., via the alignment between the second inlet opening **1212** of the cone **1200** and the inlet opening **910** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** of the gas train **900** into the flow chamber **1304** of the cone **1200** via the inlet opening **910** of the chamber **904** and the second inlet opening **1212** of the cone **1200**. When the cone **1200** is positioned in the above-described second open position, the flow chamber **1304** of the cone **1200** is also in fluid communication with the first outlet flow channel **906** of the gas train **900** (e.g., via the first outlet opening **1306** of the cone **1200** and the first outlet opening **912** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the first outlet flow channel **906** of the gas train **900** via the first outlet opening **1306** of the cone **1200** and the first outlet opening **912** of the chamber **904**.

[0121] When the cone **1200** is positioned in the above-described second open position, there is once again a lack of alignment between the second outlet opening **1216** formed in the sidewall **1202** of the cone **1200** and the second outlet opening **914** of the chamber **904**. The second outlet opening **914** of the chamber **904** is accordingly blocked and/or covered by a solid portion of the sidewall **1202** of the cone **1200**, thereby preventing the flow chamber **1304** of the cone **1200** from being in fluid communication with the second outlet flow channel **1102** of the gas train **900**, and thereby also preventing the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the second outlet flow channel **1102** of the gas train **900** via

the second outlet opening **914** of the chamber **904**.

[0122] As another example, the cone **1200** can be further rotated relative to the chamber **904** of the gas train **900** by an additional ninety degree counterclockwise relative to the second open position described above, thereby placing the cone **1200** in a third open position (e.g., one-hundred eighty degrees from the open position **2300** of FIGS. 23-25). When the cone **1200** is positioned in such a third open position, the third inlet opening **1214** formed in the sidewall **1202** of the cone **1200** is aligned with the inlet opening **910** of the chamber **904**. The flow chamber **1304** of the cone **1200** is accordingly in fluid communication with the inlet flow channel **902** of the gas train **900** (e.g., via the alignment between the third inlet opening **1214** of the cone **1200** and the inlet opening **910** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the inlet flow channel **902** of the gas train **900** into the flow chamber **1304** of the cone **1200** via the inlet opening **910** of the chamber **904** and the third inlet opening **1214** of the cone **1200**. When the cone **1200** is positioned in the above-described third open position, the flow chamber **1304** of the cone **1200** is also in fluid communication with the first outlet flow channel **906** of the gas train **900** (e.g., via the first outlet opening **1306** of the cone **1200** and the first outlet opening **912** of the chamber **904**), thereby enabling the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the first outlet flow channel **906** of the gas train **900** via the first outlet opening **1306** of the cone **1200** and the first outlet opening **912** of the chamber **904**.

[0123] When the cone **1200** is positioned in the above-described third open position, there is still a lack of alignment between the second outlet opening **1216** formed in the sidewall **1202** of the cone **1200** and the second outlet opening **914** of the chamber **904**. The second outlet opening **914** of the chamber **904** is accordingly blocked and/or covered by a solid portion of the sidewall **1202** of the cone **1200**, thereby preventing the flow chamber **1304** of the cone **1200** from being in fluid communication with the second outlet flow channel **1102** of the gas train **900**, and thereby also preventing the movement of pressurized fluid (e.g., pressurized gas) from the flow chamber **1304** of the cone **1200** into the second outlet flow channel **1102** of the gas train **900** via the second outlet opening **914** of the chamber **904**.

[0124] The gas valve **100** of FIGS. 1-25 can be implemented in a variety of grills. FIG. 26 is a perspective view of an example grill **2600** constructed in accordance with the teachings of this disclosure. FIG. 27 is an exploded view of the grill **2600** of FIG. 26. In the illustrated example of FIGS. 26 and 27, the grill **2600** is structured as a flat top gas grill. In this regard, the grill **2600** of FIGS. 26 and 27 includes an example cookbox **2602** and an example griddle **2604**. In other examples, the grill **2600** can instead be configured as a different type of gas grill. For example, the grill **2600** can instead be configured as a gas grill including a cookbox that supports and/or carries one or more cooking grate(s). The grill **2600** of FIGS. 26 and 27 is configured to include at least one instance of the above-described gas valve **100**, with said implementation being further described herein.

[0125] In the illustrated example of FIGS. 26 and 27, the cookbox **2602** includes an example front wall **2702**, an example rear wall **2704**, an example right sidewall **2706**, and an example left sidewall **2708**. The rear wall **2704** of the cookbox **2602** is located opposite the front wall **2702** of the cookbox **2602**. The right sidewall **2706** of the cookbox **2602** extends between the front wall **2702** and the rear wall **2704** of the cookbox **2602**. The left sidewall **2708** of the cookbox **2602** is located opposite the right sidewall **2706** of the cookbox **2602** and extends between the front wall **2702** and the rear wall **2704** of the cookbox **2602**. As shown in FIGS. 26 and 27, the front wall **2702**, the rear wall **2704**, the right sidewall **2706**, and the left sidewall **2708** of the cookbox **2602** define a cavity in which one or more structural component(s) (e.g., one or more burner tube(s)) of the grill **2600** can be disposed.

[0126] The griddle **2604** of the grill **2600** is configured to be disposed on and/or above the cookbox **2602** of the grill **2600**. In the illustrated example of FIGS. 26 and 27, the griddle **2604** includes an example base **2710** having an example flat top cooking surface **2712** and an underside located

opposite the flat top cooking surface **2712**. In the illustrated example of FIGS. **26** and **27**, the flat top cooking surface **2712** is a continuous, substantially flat and/or substantially planar surface that is free of openings aside from an example waste disposal opening **2606** formed therein. The flat top cooking surface **2712** is configured to support a variety of liquid, semi-liquid, and/or solid food items during a variety of cooking processes that may be performed on the griddle **2604**. In the illustrated example of FIGS. **26** and **27**, the flat top cooking surface **2712** and/or, more generally, the base **2710** of the griddle **2604** has a generally rectangular profile. In other examples, the flat top cooking surface **2712** and/or, more generally, the base **2710** of the griddle **2604** can instead have a non-rectangular profile (e.g., a circular profile, an oval profile, a triangular profile, a trapezoidal profile, etc.).

[0127] The griddle **2604** of FIGS. **26** and **27** further includes an example front lip **2714**, and an example rear lip **2716**, an example right side lip **2718**, and an example left side lip **2720**. The rear lip **2716** of the griddle **2604** is located opposite the front lip **2714** of the griddle **2604**. The right side lip **2718** of the griddle **2604** extends between the front lip **2714** and the rear lip **2716** of the griddle **2604**. The left side lip **2720** of the griddle **2604** is located opposite the right side lip **2718** of the griddle **2604** and extends between the front lip **2714** and the rear lip **2716** of the griddle **2604**. The front lip **2714**, the rear lip **2716**, the right side lip **2718**, and the left side lip **2720** of the griddle **2604** extend upwardly from the base **2710** of the griddle **2604** to provide vertical boundaries configured to prevent food items from sliding off the flat top cooking surface **2712** of the griddle **2604** during one or more cooking operation(s) performed thereon.

[0128] The grill **2600** of FIGS. **26** and **27** further includes a plurality of burner tubes configured to be supported by, carried by, and/or disposed within the cookbox **2602** of the grill **2600**. As shown in FIG. **27**, the plurality of burner tubes includes an example first burner tube **2722**, an example second burner tube **2724**, and an example third burner tube **2726** (e.g., a total of three burner tubes) configured to be laterally spaced apart from one another and arranged in a front-to-rear orientation within the cookbox **2602**, with the first burner tube **2722**, the second burner tube **2724**, and the third burner tube **2726** being of a substantially identical construction relative to one another. In other examples, the plurality of burner tubes can instead include a different number (e.g., two, four, five, etc.) of burner tubes, and the construction of one or more of the burner tubes may differ from that of the first burner tube **2722**, the second burner tube **2724**, and/or the third burner tube **2726** shown in FIG. **27**. The first burner tube **2722** of the grill **2600** is further described below in connection with FIGS. **28-34**.

[0129] In the illustrated example of FIGS. **26** and **27**, the burner tubes (e.g., the first burner tube **2722**, the second burner tube **2724**, and the third burner tube **2726**) form part of a gas train that further includes an example fuel source **2608**, an example regulator assembly **2610**, an example manifold **2728**, and a plurality of control valves corresponding in number to the plurality of burner tubes. In this regard, the plurality of control valves of the grill **2600** as shown in FIG. **27** includes an example first control valve **2730** associated with the first burner tube **2722**, an example second control valve **2732** associated with the second burner tube **2724**, and an example third control valve **2734** associated with the third burner tube **2726**. In the illustrated example of FIGS. **26** and **27**, the first control valve **2730**, the second control valve **2732**, and the third control valve **2734** of the grill **2600** are respectively implemented by separate instances of the gas valve **100** of FIGS. **1-11** and **20-25** described above.

[0130] In the illustrated example of FIGS. **26** and **27**, the fuel source **2608** is implemented as a fuel tank (e.g., a propane tank) containing combustible gas. In other examples, the fuel source **2608** can instead be implemented as a piped (e.g., household) natural gas line that provides an accessible flow of combustible gas. The regulator assembly **2610** is operatively positioned between the fuel source **2608** and the manifold **2728** such that a supply of pressurized combustible gas provided via the fuel source **2608** flows through the regulator assembly **2610** and into the manifold **2728**. The first control valve **2730** is operatively positioned between the manifold **2728** and the first burner

tube **2722** such that pressurized combustible gas received at the manifold **2728** can be selectively supplied to the first burner tube **2722** via the first control valve **2730**. The second control valve **2732** is operatively positioned between the manifold **2728** and the second burner tube **2724** such that pressurized combustible gas received at the manifold **2828** can be selectively supplied to the second burner tube **2724** via the second control valve **2732**. The third control valve **2734** is operatively positioned between the manifold **2728** and the third burner tube **2726** such that pressurized combustible gas received at the manifold **2728** can be selectively supplied to the third burner tube **2726** via the third control valve **2734**.

[0131] In the illustrated example of FIGS. **26** and **27**, the manifold **2728**, the first control valve **2730**, the second control valve **2732**, and the third control valve **2734** are at least partially covered and/or concealed by an example control panel **2612** that is coupled to and/or located along the front of the cookbox **2602**. The grill **2600** further includes a plurality of control knobs mounted and/or located along the front face of the control panel **2612**, with the plurality of control knobs corresponding in number to the plurality of control valves and/or the number of burner tubes. In this regard, the plurality of control valves of the grill **2600** as shown in FIGS. **26** and **27** includes an example first control knob **2614** associated with the first control valve **2730** and/or the first burner tube **2722**, an example second control knob **2616** associated with the second control valve **2732** and/or the second burner tube **2724**, and an example third control knob **2618** associated with the third control valve **2734** and/or the third burner tube **2726**. Each control knob is mechanically coupled to its corresponding control valve such that movement (e.g., rotation) of the control knob changes the extent to which an adjustable flow control member of the corresponding control valve enables pressurized combustible gas to flow through the corresponding control valve into the corresponding burner tube. For example, the first control knob **2614** is mechanically coupled to the first control valve **2730** such that movement (e.g., rotation) of the first control knob **2614** changes the extent to which an adjustable flow control member of the first control valve **2730** enables pressurized combustible gas to flow through the first control valve **2730** into the first burner tube **2722**.

[0132] The grill **2600** of FIGS. **26** and **27** further includes an example frame **2620**. The frame **2620** can be configured from any number and any type of structural components arranged in any manner that facilitates supporting the cookbox **2602** above an underlying ground surface when the grill **2600** is in use. In the illustrated example of FIGS. **26** and **27**, the frame **2620** includes an example right side support panel **2622** and an example left side support panel **2624**, each of which is configured to support the cookbox **2602**. As shown in FIGS. **26** and **27**, the right side support panel **2622** and the left side support panel **2624** are spaced apart from one another, are oriented vertically, and are coupled (e.g., via one or more fastener(s)) to the cookbox **2602** in a fixed manner. The frame **2620** further includes an example upper shelf **2626** and an example lower shelf **2628**, each of which is configured to support one or more item(s) at a location below the cookbox **2602** of the grill **2600**. As shown in FIGS. **26** and **27**, the upper shelf **2626** and the lower shelf **2628** are spaced apart from one another, are oriented horizontally, and are coupled (e.g., via one or more fastener(s)) to the right side support panel **2622** and/or the left side support panel **2624** in a fixed manner. In other examples, the frame **2620** can instead include one or more foldable, slidable, and/or telescoping support member(s) (e.g., leg(s), panel(s), etc.) that facilitate collapsing and/or otherwise modifying the frame **2620** of the grill **2600** when the grill **2600** is not in use.

[0133] The grill **2600** of FIGS. **26** and **27** further includes an example right side accessory support frame **2630** and an example left side accessory support frame **2632**. The right side accessory support frame **2630** is coupled (e.g., via one or more fastener(s)) to the right side support panel **2622** of the frame **2620**, and/or to a right sidewall of the cookbox **2602**. Conversely, the left side accessory support frame **2632** is coupled (e.g., via one or more fastener(s)) to the left side support panel **2624** of the frame **2620**, and/or to a left sidewall of the cookbox **2602**. The right side accessory support frame **2630** and the left side accessory support frame **2632** of the grill **2600** are

respectively configured to support one or more insertable accessories and/or one or more snap fit accessories at a location to the side (e.g., the right side or the left side) of the cookbox **2602** of the grill **2600**.

[0134] The grill **2600** of FIGS. **26** and **27** further includes an example lid **2634** configured to cover and/or enclose the griddle **2604** of the grill **2600** when the lid **2634** is in a closed position. The lid **2634** is movable relative to the cookbox **2602**, the griddle **2604**, and/or the frame **2620** of the grill **2600** between a closed position in which the flat top cooking surface **2712** of the griddle **2604** is covered, and an open position in which the flat top cooking surface **2712** of the griddle **2604** is exposed (e.g., as shown in FIG. **26**). In the illustrated example of FIGS. **26** and **27**, the lid **2634** is pivotally coupled to the cookbox **2602** and/or the frame **2620** of the grill **2600** via one or more hinge(s). In other examples, the lid **2634** can instead be removably positioned on the cookbox **2602**, the griddle **2604**, and/or the frame **2620** of the grill **2600** without there being any direct mechanical coupling between the lid **2634** on the one hand and the cookbox **2602**, the griddle **2604**, and/or the frame **2620** on the other hand. Movement of the lid **2634** of the grill **2600** between the closed position and the open position can be facilitated via user interaction with an example handle **2636** that is coupled (e.g., via one or more fastener(s)) to the lid **2634**.

[0135] FIG. **28** is a perspective view showing the first burner tube **2722** of the grill **2600** of FIGS. **26** and **27** in isolation. FIG. **29** is a top view of the first burner tube **2722** of FIG. **28**. FIG. **30** is a bottom view of the first burner tube **2722** of FIGS. **28** and **29**. FIG. **31** is a right side view of the first burner tube **2722** of FIGS. **28-30**. FIG. **32** is a cross-sectional view of the first burner tube **2722** of FIGS. **28-31** taken along section C-C of FIG. **29**. The construction of the second burner tube **2724** and the third burner tube **2726** of the grill **2600** of FIGS. **26** and **27** is substantially identical to that of the first burner tube **2722** as described herein in connection with FIGS. **28-32**.

[0136] As shown in FIGS. **28-32**, the first burner tube **2722** includes an example first end **2802** (e.g., a front end), an example second end **2804** (e.g., a rear end), an example outer wall **2806** extending between the first end **2802** and the second end **2804**, and an example central axis **2808** extending between the first end **2802** and the second end **2804**. The first end **2802** of the first burner tube **2722** is partially open, with the first end **2802** being configured to receive an outlet of a control valve (e.g., the outlet of the first control valve **2730** as implemented via the first outlet **208** of the gas valve **100**) located proximate the front wall **2702** of the cookbox **2602**. The second end **2804** of the first burner tube **2722** is closed, with the second end **2804** being configured to be coupled to a mounting flange located proximate the rear wall **2704** of the cookbox **2602**. The first burner tube **2722** is accordingly configured to be arranged in a front-to-rear orientation when disposed in the cookbox **2602**. In the illustrated example of FIGS. **28-32**, the first burner tube **2722** is structured as a linear burner tube having a circular cross-sectional profile. In other example, the first burner tube **2722** can instead be structured as a linear burner tube having a different cross-sectional profile (e.g., a rectangular cross-sectional profile, an oval-shaped cross-sectional profile, a triangular cross-sectional profile, a trapezoidal cross-sectional profile, etc.). In still other examples, the first burner tube **2722** can instead be structured as a non-linear burner tube having one or more bend(s) and/or curve(s) formed therein.

[0137] The first burner tube **2722** of FIGS. **28-32** includes a plurality of ports formed in and extending through the outer wall **2806** of the first burner tube **2722**. As shown in FIGS. **28-32**, the plurality of ports includes an example first row of ports **2810**, an example second row of ports **2812**, an example third row of ports **2814**, and an example fourth row of ports **2816**. The first row of ports **2810** is arranged parallel to the central axis **2808**. The second row of ports **2812** is also arranged parallel to the central axis **2808**, with the second row of ports **2812** being spaced apart from (e.g., angularly displaced from) the first row of ports **2810** about the perimeter (e.g., about the circumference) of the outer wall **2806** of the first burner tube **2722**. The third row of ports **2814** is arranged perpendicular to the central axis **2808**. The fourth row of ports **2816** is also arranged perpendicular to the central axis **2808**, with the fourth row of ports **2816** being spaced apart from

(e.g., longitudinally displaced from) the third row of ports **2814** along the central axis **2808** of the first burner tube **2722**. In the illustrated example of FIGS. **28-32**, the third row of ports **2814** is located proximate the first end **2802** (e.g., the front end) of the first burner tube **2722**, and the fourth row of ports **2816** is located proximate the second end **2804** (e.g., the rear end) of the first burner tube **2722**. The first burner tube **2722** can include additional ports and/or additional rows of ports relative to those described above. For example, as shown in FIGS. **28-32**, the first burner tube **2722** further includes an example fifth row of ports **2818** arranged perpendicular to the central axis **2808**, with the fifth row of ports **2818** being spaced apart from and proximate to the third row of ports **2814**.

[0138] As shown in FIGS. **28-32**, the outer wall **2806** of the first burner tube **2722** has a circular cross-sectional profile. The second row of ports **2812** is angularly displaced from the first row of ports **2810** about the circumference of the outer wall **2806** by an example angle **3202** measuring approximately 120.0 degrees. In other examples, the angle **3202** at which the second row of ports **2812** is angularly displaced from the first row of ports **2810** about the circumference of the outer wall **2806** can be between 90.0 and 150.0 degrees. Satisfaction of the above-described range of the angle **3202** at which the second row of ports **2812** is angularly displaced from the first row of ports **2810** about the circumference of the outer wall **2806** advantageously improves the operating efficiency of the cooking engine of the grill **2600**, and also improves the heat distribution properties associated with the griddle **2604** of the grill **2600**. In this regard, satisfaction of the above-described range of the angle **3202** at which the second row of ports **2812** is angularly displaced from the first row of ports **2810** about the circumference of the outer wall **2806** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2712** of the griddle **2604** during cooking operations performed thereon.

[0139] The first burner tube **2722** of FIGS. **28-32** is further configured such that the third row of ports **2814** of the first burner tube **2722** is spaced apart from the first end **2802** (e.g., the front end) of the first burner tube **2722** by an example distance **2902** of approximately 4.4 inches. In other examples, the distance **2902** between the third row of ports **2814** and the first end **2802** of the first burner tube **2722** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **2902** between the third row of ports **2814** and the first end **2802** of the first burner tube **2722** advantageously improves the operating efficiency of the cooking engine of the grill **2600**, and also improves the heat distribution properties associated with the griddle **2604** of the grill **2600**. In this regard, satisfaction of the above-described range of the distance **2902** between the third row of ports **2814** and the first end **2802** of the first burner tube **2722** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2712** of the griddle **2604** during cooking operations performed thereon.

[0140] The first burner tube **2722** of FIGS. **28-32** is further configured such that the fourth row of ports **2816** of the first burner tube **2722** is spaced apart from the second end **2804** (e.g., the rear end) of the first burner tube **2722** by an example distance **2904** of approximately 1.8 inches. In other examples, the distance **2904** between the fourth row of ports **2816** and the second end **2804** of the first burner tube **2722** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **2904** between the fourth row of ports **2816** and the second end **2804** of the first burner tube **2722** advantageously improves the operating efficiency of the cooking engine of the grill **2600**, and also improves the heat distribution properties associated with the griddle **2604** of the grill **2600**. In this regard, satisfaction of the above-described range of the distance **2904** between the fourth row of ports **2816** and the second end **2804** of the first burner tube **2722** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2712** of the griddle **2604** during cooking operations performed thereon.

[0141] In the illustrated example of FIGS. **28-32**, the first burner tube **2722** has a length of approximately 18.6 inches measured along the central axis **2808** from the first end **2802** (e.g., the front end) to the second end **2804** (e.g., the rear end) of the first burner tube **2722**. In other

examples, the first burner tube **2722** can instead have a length that is substantially greater than or substantially less than 18.6 inches. In the illustrated example of FIGS. **28-32**, the outer wall **1806** of the first burner tube **2722** has a diameter of approximately 1.0 inches measured across the circular cross-sectional profile of the outer wall **1806**. In other examples, the first burner tube **2722** can instead have a diameter that is substantially greater than or substantially less than 1.0 inches. In the illustrated example of FIGS. **28-32**, each one of the ports formed in and extending through the outer wall **2806** of the first burner tube **2722** has a diameter ranging between approximately 0.07 inches and 0.08 inches measured across the circular outlet opening of the port. In other examples, one or more of the ports formed in and extending through the outer wall **1806** of the first burner tube **2722** can instead have a diameter that is substantially less than 0.07 inches or substantially greater than 0.08 inches.

[0142] FIG. **33** is a perspective view of a portion of an example burner assembly **3300** of the grill **2600** of FIGS. **26** and **27**. FIG. **34** is a side view of the burner assembly **3300** of FIG. **33**. The illustrated portion of the burner assembly **3300** of FIGS. **33** and **34** includes the manifold **2728**, the first control valve **2730**, the first burner tube **2722**, and the first control knob **2614** of the grill **2600**. Although not shown in FIGS. **33** and **34**, the burner assembly **3300** can additionally include the second control valve **2732**, the second burner tube **2724**, the second control knob **2616**, the third control valve **2734**, the third burner tube **2726**, and the third control knob **2618** of the grill **2600**. The first control valve **2730** of FIGS. **33** and **34** is implemented by an instance of the gas valve **100** of FIGS. **1-25** described above.

[0143] In the illustrated example of FIGS. **33** and **34**, the inlet **202** and/or the first end **212** of the inlet conduit **204** of the gas valve **100** is/are configured to be fluidically coupled to the manifold **2728** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the manifold **2728** is able to flow from an outlet and/or an opening of the manifold **2728** into the inlet **202** and/or into the first end **212** of the inlet conduit **204**. The second end **218** of the first outlet conduit **108** of the gas valve **100** and/or the first outlet **208** of the gas valve **100** is/are configured to be fluidically coupled to the open first end **2802** of the first burner tube **2722** such that pressurized fluid (e.g., pressurized gas) present in and/or flowing through the first outlet conduit **108** and/or the first outlet **208** is able to flow from the first outlet conduit **108** and/or the first outlet **208** into the open first end **2802** of the first burner tube **2722**. The first control knob **2614** is mechanically coupled to the stem **104** of the gas valve **100** such that rotation of the first control knob **2614** causes a corresponding rotation of the stem **104** about the axis of rotation **1006** of the stem **104**, which is coaxially aligned with the central axis of the chamber **904** as well as the central axis **1402** of the cone **1200** of the gas valve **100**.

[0144] In the illustrated example of FIGS. **33** and **34**, the central axis **2808** of the first burner tube **2722** is coaxially aligned with the central axis **1014** of the fourth linear segment **922** of the first outlet flow channel **906** of the gas valve **100**. Furthermore, the central axis of the first burner tube **2722** is parallel to and offset from the axis of rotation **1006** of the stem **104**, parallel to and offset from the central axis of the chamber **904**, and/or parallel to and offset from the central axis **1402** of the cone **1200** of the gas valve **100**. For example, as shown in FIGS. **33** and **34**, the central axis of the first burner tube **2722** is parallel to and located above the axis of rotation **1006** of the stem **104**, parallel to and located above the central axis of the chamber **904**, and parallel to and located above the central axis **1402** of the cone **1200** of the gas valve **100**.

[0145] As described above in connection with FIGS. **9-11**, the serpentine (e.g., S-shaped) portion of the gas train **900** of FIGS. **9-11** advantageously reduces (e.g., minimizes) the front-to-rear distance between the inlet **202** of the gas valve **100** and the first outlet **208** of the gas valve **100**, and/or the front-to-rear distance between the stem **104** of the gas valve **100** and the first outlet **208** of the gas valve **100**. As shown in FIGS. **33** and **34**, the serpentine (e.g., S-shaped) portion of the gas train **900** of FIGS. **9-11** also advantageously reduces (e.g., minimizes) the front-to-rear distance between an example rear end **3302** of the first control knob **2614** and the first end **2802** (e.g., the front end)

of the first burner tube **2722**. In the illustrated example of FIGS. **33** and **34**, the example distance **3402** between the rear end **3302** of the first control knob **2614** and the first end **2802** (e.g., the front end) of the first burner tube **2722** is approximately 1.8 inches. In other examples, the distance **3402** between the rear end **3302** of the first control knob **2614** and the first end **2802** (e.g., the front end) of the first burner tube **2722** can be between 0.5 and 3.0 inches. The improved space efficiency associated with this/these reduced dimension(s) of the burner assembly **3300** advantageously enables the gas valve **100** of the burner assembly **3300** to be disposed within a correspondingly smaller (e.g., reduced depth) control panel chamber of the grill **2600**, and also advantageously enables the first burner tube **2722** of the grill **2600** to be located closer to an interior of the front wall **2702** of the cookbox **2602** of the grill **2600**, as further described herein.

[0146] FIG. **35** is a front view showing the grill **2600** of FIGS. **26** and **27**, with the frame **2620**, the lid **2634**, and certain other structural features of the grill **2600** omitted for enhanced viewability. FIG. **36** is a top view of the grill **2600** of FIG. **35**. FIG. **37** is a cross-sectional view of the grill **2600** of FIGS. **35** and **36** taken along section D-D of FIG. **36**. FIG. **38** is a cross-sectional view of the grill **2600** of FIGS. **35-37** taken along section E-E of FIG. **35**. As shown in FIGS. **35-38**, the griddle **2604** of the grill **2600** is disposed on or above the cookbox **2602** of the grill **2600**. As further shown in FIGS. **35-38**, the grill **2600** includes an example control panel chamber **3702** located between the control panel **2612** of the grill **2600** and the front wall **2702** of the cookbox **2602** of the grill **2600**.

[0147] As discussed above in connection with FIGS. **33** and **34** and further shown in FIGS. **35-38**, the distance **3402** between the rear end **3302** of the first control knob **2614** of the grill **2600** and the first end **2802** (e.g., the front end) of the first burner tube **2722** of the grill **2600** is approximately 1.8 inches. In other examples, the distance **3402** between the rear end **3302** of the first control knob **2614** of the grill **2600** and the first end **2802** (e.g., the front end) of the first burner tube **2722** of the grill **2600** can be between 0.5 and 3.0 inches. The improved space efficiency associated with this/these reduced dimension(s) of the burner assembly **3300** advantageously enables the control panel chamber **3702** of the grill **2600** to be configured with a correspondingly reduced depth. For example, as shown in FIGS. **35-38**, the control panel **2612** of the grill **2600** is spaced apart from an example exterior surface **3704** of the front wall **2702** of the cookbox **2602** of the grill **2600** by an example distance **3706** of approximately 2.5 inches measured depthwise from the control panel **2612** to the exterior surface **3704** of the front wall **2702** of the cookbox **2602**. In other examples, the distance **3706** at which the control panel **2612** of the grill **2600** is spaced apart from the exterior surface **3704** of the front wall **2702** of the cookbox **2602** of the grill **2600** can be between 1.0 and 4.0 inches. As another example, as shown in FIGS. **35-38**, the control panel **2612** of the grill **2600** is spaced apart from an example interior surface **3708** of the front wall **2702** of the cookbox **2602** of the grill **2600** by an example distance **3710** of approximately 4.4 inches measured depthwise from the control panel **2612** to the interior surface **3708** of the front wall **2702** of the cookbox **2602**. In other examples, the distance **3710** at which the control panel **2612** of the grill **2600** is spaced apart from the interior surface **3708** of the front wall **2702** of the cookbox **2602** of the grill **2600** can be between 3.0 and 6.0 inches.

[0148] In the illustrated example of FIGS. **35-38**, the third row of ports **2814** formed on the outer wall **2806** of the first burner tube **2722** is positioned inwardly from the interior surface **3708** of the front wall **2702** of the cookbox **2602** by an example distance **3712** of approximately 1.5 inches measured depthwise (e.g., along the central axis **2808** of the first burner tube **2722**) from the interior surface **3708** of the front wall **2702** of the cookbox **2602** to the third row of ports **2814** of the first burner tube **2722**. In other examples, the distance **3712** at which the third row of ports **2814** of the first burner tube **2722** is positioned inwardly from the interior surface **3708** of the front wall **2702** of the cookbox **2602** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **3712** at which the third row of ports **2814** of the first burner tube **2722** is positioned inwardly from the interior surface **3708** of the front wall **2702** of the cookbox

2602 advantageously improves the operating efficiency of the cooking engine of the grill **2600**, and also improves the heat distribution properties associated with the griddle **2604** of the grill **2600**. In this regard, satisfaction of the above-described range of the distance **3712** at which the third row of ports **2814** of the first burner tube **2722** is positioned inwardly from the interior surface **3708** of the front wall **2702** of the cookbox **2602** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2712** of the griddle **2604** during cooking operations performed thereon.

[0149] In the illustrated example of FIGS. 35-38, the third row of ports **2814** formed on the outer wall **2806** of the first burner tube **2722** is positioned inwardly from the front lip **2714** of the griddle **2604** by an example distance **3714** of approximately 3.3 inches measured depthwise (e.g., along the central axis **2808** of the first burner tube **2722**) from the front lip **2714** of the griddle **2604** to the third row of ports **2814** of the first burner tube **2722**. In other examples, the distance **3714** at which the third row of ports **2814** of the first burner tube **2722** is positioned inwardly from the front lip **2714** of the griddle **2604** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **3714** at which the third row of ports **2814** of the first burner tube **2722** is positioned inwardly from the front lip **2714** of the griddle **2604** advantageously improves the operating efficiency of the cooking engine of the grill **2600**, and also improves the heat distribution properties associated with the griddle **2604** of the grill **2600**. In this regard, satisfaction of the above-described range of the distance **3714** at which the third row of ports **2814** of the first burner tube **2722** is positioned inwardly from the front lip **2714** of the griddle **2604** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2712** of the griddle **2604** during cooking operations performed thereon.

[0150] The following paragraphs provide various examples in relation to the disclosed space efficient gas valves for grills.

[0151] Example 1 includes a gas valve. In Example 1, the gas valve includes a chamber, a flow control member, an inlet conduit, an outlet conduit, and a stem. The flow control member is disposed within the chamber. The flow control member is rotatable within the chamber between an open position and a closed position. The inlet conduit extends downwardly from the chamber. The inlet conduit includes an inlet flow channel having an inlet. The outlet conduit extends upwardly from the chamber. The outlet conduit includes an outlet flow channel having an outlet. The stem is operatively coupled to the flow control member such that rotation of the stem about an axis of rotation of the stem causes a corresponding rotation of the flow control member within the chamber about an axis of rotation of the flow control member. The outlet of the outlet flow channel is located above the axis of rotation of the of the flow control member.

[0152] Example 2 includes the gas valve of Example 1. In Example 2, the inlet conduit is configured to be coupled to a manifold of a grill such that the inlet is in fluid communication with the manifold. In Example 2, the outlet conduit is configured to be coupled to a burner tube of the grill such that the outlet is in fluid communication with the burner tube.

[0153] Example 3 includes the gas valve of Example 1. In Example 3, the axis of rotation of the flow control member is coaxially aligned with the axis of rotation of the stem.

[0154] Example 4 includes the gas valve of Example 1. In Example 4, the chamber includes an inlet opening and an outlet opening, and the flow control member includes a flow chamber having an inlet opening and an outlet opening.

[0155] Example 5 includes the gas valve of Example 4. In Example 5, the inlet flow channel of the inlet conduit is in fluid communication with the flow chamber of the flow control member via the inlet opening of the chamber and the inlet opening of the flow chamber when the flow control member is in the open position. In Example 5, the inlet flow channel of the inlet conduit is not in fluid communication with the flow chamber of the flow control member when the flow control member is in the closed position.

[0156] Example 6 includes the gas valve of Example 5. In Example 6, the outlet flow channel of

the outlet conduit is in fluid communication with the flow chamber of the flow control member via the outlet opening of the flow control member and the outlet opening of the chamber when the flow control member is in the open position.

[0157] Example 7 includes the gas valve of Example 6. In Example 7, the outlet flow channel of the outlet conduit is also in fluid communication with the flow chamber of the flow control member via the outlet opening of the flow control member and the outlet opening of the chamber when the flow control member is in the closed position.

[0158] Example 8 includes the gas valve of Example 6. In Example 8, the outlet opening of the chamber is a first outlet opening of the chamber and the chamber further includes a second outlet opening. In Example 8, the outlet opening of the flow chamber is a first outlet opening of the flow chamber and the flow chamber further includes a second outlet opening. In Example 8, the outlet conduit is a first outlet conduit of the gas valve, the outlet flow channel is a first outlet flow channel, and the outlet is a first outlet. In Example 8, the gas valve further includes a second outlet conduit including a second outlet flow channel having a second outlet. In Example 8, the second outlet flow channel of the second outlet conduit is in fluid communication with the flow chamber of the flow control member via the second outlet opening of the flow chamber and the second outlet opening of the chamber when the flow control member is in the open position. In Example 8, the second outlet flow channel of the second outlet conduit is not in fluid communication with the flow chamber of the flow control member when the flow control member is in the closed position.

[0159] Example 9 includes the gas valve of Example 8. In Example 9, the second outlet conduit is configured to be coupled to an ignition conduit of a grill such that the second outlet is in fluid communication with the ignition conduit.

[0160] Example 10 includes the gas valve of Example 1. In Example 10, a gas train of the gas valve includes an S-shaped portion defined by the chamber and the outlet flow channel.

[0161] Example 11 includes the gas valve of Example 10. In Example 11, the outlet flow channel includes a first segment extending upwardly from and being in fluid communication with the chamber, a second segment extending forwardly from and being in fluid communication with the first segment, a third segment extending upwardly from and being in fluid communication with the second segment, and a fourth segment extending rearwardly from and being in fluid communication with the third segment. In Example 11, the fourth segment includes the outlet. In Example 11, the S-shaped portion of the gas train includes the chamber, the first segment, the second segment, the third segment, and the fourth segment.

[0162] Example 12 includes the gas valve of Example 11. In Example 12, the first segment, the second segment, the third segment, and the fourth segment are linear.

[0163] Example 13 includes the gas valve of Example 12. In Example 13, a central axis of the chamber, a central axis of the second segment, and a central axis of the fourth segment are parallel to one another and horizontally oriented. In Example 13, a central axis of the first segment and a central axis of the third segment are parallel to one another and vertically oriented.

[0164] Example 14 includes the gas valve of Example 13. In Example 14, the central axis of the second segment is located above the central axis of the chamber, the central axis of the fourth segment is located above the central axis of the second segment, and the central axis of the third segment is located forward of the central axis of the first segment.

[0165] Example 15 includes the gas valve of Example 14. In Example 15, the central axis of the third segment is parallel to and located forward of a central axis of the inlet flow channel.

[0166] Example 16 includes a grill. In Example 16, the grill includes a gas valve, a manifold, a burner tube, and a control knob. In Example 16, the gas valve includes a chamber, a flow control member, an inlet conduit, an outlet conduit, and a stem. The flow control member is disposed within the chamber. The flow control member is rotatable within the chamber between an open position and a closed position. The inlet conduit extends downwardly from the chamber. The inlet conduit includes an inlet flow channel having an inlet. The outlet conduit extends upwardly from

the chamber. The outlet conduit includes an outlet flow channel having an outlet. The stem is operatively coupled to the flow control member such that rotation of the stem about an axis of rotation of the stem causes a corresponding rotation of the flow control member within the chamber about an axis of rotation of the flow control member. The axis of rotation of the flow control member is coaxially aligned with the axis of rotation of the stem. The outlet of the outlet flow channel is located above the axis of rotation of the of the flow control member. The manifold is coupled to the inlet conduit of the gas valve such that the inlet is in fluid communication with the manifold. The burner tube is coupled to the outlet conduit of the gas valve such that the outlet is in fluid communication with the burner tube. The control knob is operatively coupled to the stem such that rotation of the control knob causes a corresponding rotation of the stem.

[0167] Example 17 includes the grill of Example 16. In Example 17, a gas train of the gas valve includes an S-shaped portion defined by the chamber and the outlet flow channel.

[0168] Example 18 includes the grill of Example 17. In Example 18, the outlet flow channel includes a first segment extending upwardly from and being in fluid communication with the chamber, a second segment extending forwardly from and being in fluid communication with the first segment, a third segment extending upwardly from and being in fluid communication with the second segment, and a fourth segment extending rearwardly from and being in fluid communication with the third segment. In Example 18, the fourth segment includes the outlet. In Example 18, the S-shaped portion of the gas train includes the chamber, the first segment, the second segment, the third segment, and the fourth segment.

[0169] Example 19 includes the grill of Example 18. In Example 19, the first segment, the second segment, the third segment, and the fourth segment are linear. In Example 19, a central axis of the chamber, a central axis of the second segment, and a central axis of the fourth segment are parallel to one another and horizontally oriented. In Example 19, a central axis of the first segment and a central axis of the third segment are parallel to one another and vertically oriented.

[0170] In Example 20, the central axis of the second segment is located above the central axis of the chamber, the central axis of the fourth segment is located above the central axis of the second segment, the central axis of the third segment is located forward of the central axis of the first segment, and the central axis of the third segment is parallel to and located forward of a central axis of the inlet flow channel.

[0171] Although certain example apparatus, systems, methods, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all apparatus, systems, methods, and articles of manufacture fairly falling within the scope of the claims of this patent.

[0172] The following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

Claims

1. A gas valve, comprising: a chamber; a flow control member disposed within the chamber, the flow control member being rotatable within the chamber between an open position and a closed position; an inlet conduit extending downwardly from the chamber, the inlet conduit including an inlet flow channel having an inlet; an outlet conduit extending upwardly from the chamber, the outlet conduit including an outlet flow channel having an outlet; and a stem operatively coupled to the flow control member such that rotation of the stem about an axis of rotation of the stem causes a corresponding rotation of the flow control member within the chamber about an axis of rotation of the flow control member, wherein the outlet of the outlet flow channel is located above the axis of rotation of the of the flow control member.
2. The gas valve of claim 1, wherein the inlet conduit is configured to be coupled to a manifold of a grill such that the inlet is in fluid communication with the manifold, and wherein the outlet conduit

is configured to be coupled to a burner tube of the grill such that the outlet is in fluid communication with the burner tube.

3. The gas valve of claim 1, wherein the axis of rotation of the flow control member is coaxially aligned with the axis of rotation of the stem.

4. The gas valve of claim 1, wherein the chamber includes an inlet opening and an outlet opening, and the flow control member includes a flow chamber having an inlet opening and an outlet opening.

5. The gas valve of claim 4, wherein the inlet flow channel of the inlet conduit is in fluid communication with the flow chamber of the flow control member via the inlet opening of the chamber and the inlet opening of the flow chamber when the flow control member is in the open position, and wherein the inlet flow channel of the inlet conduit is not in fluid communication with the flow chamber of the flow control member when the flow control member is in the closed position.

6. The gas valve of claim 5, wherein the outlet flow channel of the outlet conduit is in fluid communication with the flow chamber of the flow control member via the outlet opening of the flow control member and the outlet opening of the chamber when the flow control member is in the open position.

7. The gas valve of claim 6, wherein the outlet flow channel of the outlet conduit is also in fluid communication with the flow chamber of the flow control member via the outlet opening of the flow control member and the outlet opening of the chamber when the flow control member is in the closed position.

8. The gas valve of claim 6, wherein the outlet opening of the chamber is a first outlet opening of the chamber and the chamber further includes a second outlet opening, wherein the outlet opening of the flow chamber is a first outlet opening of the flow chamber and the flow chamber further includes a second outlet opening, wherein the outlet conduit is a first outlet conduit of the gas valve, the outlet flow channel is a first outlet flow channel, and the outlet is a first outlet, wherein the gas valve further comprises a second outlet conduit including a second outlet flow channel having a second outlet, wherein the second outlet flow channel of the second outlet conduit is in fluid communication with the flow chamber of the flow control member via the second outlet opening of the flow chamber and the second outlet opening of the chamber when the flow control member is in the open position, and wherein the second outlet flow channel of the second outlet conduit is not in fluid communication with the flow chamber of the flow control member when the flow control member is in the closed position.

9. The gas valve of claim 8, wherein the second outlet conduit is configured to be coupled to an ignition conduit of a grill such that the second outlet is in fluid communication with the ignition conduit.

10. The gas valve of claim 1, wherein a gas train of the gas valve includes an S-shaped portion defined by the chamber and the outlet flow channel.

11. The gas valve of claim 10, wherein the outlet flow channel includes a first segment extending upwardly from and being in fluid communication with the chamber, a second segment extending forwardly from and being in fluid communication with the first segment, a third segment extending upwardly from and being in fluid communication with the second segment, and a fourth segment extending rearwardly from and being in fluid communication with the third segment, wherein the fourth segment includes the outlet, wherein the S-shaped portion of the gas train includes the chamber, the first segment, the second segment, the third segment, and the fourth segment.

12. The gas valve of claim 11, wherein the first segment, the second segment, the third segment, and the fourth segment are linear.

13. The gas valve of claim 12, wherein a central axis of the chamber, a central axis of the second segment, and a central axis of the fourth segment are parallel to one another and horizontally oriented, and wherein a central axis of the first segment and a central axis of the third segment are

parallel to one another and vertically oriented.

14. The gas valve of claim 13, wherein the central axis of the second segment is located above the central axis of the chamber, the central axis of the fourth segment is located above the central axis of the second segment, and the central axis of the third segment is located forward of the central axis of the first segment.

15. The gas valve of claim 14, wherein the central axis of the third segment is parallel to and located forward of a central axis of the inlet flow channel.

16. A grill, comprising: a gas valve including: a chamber; a flow control member disposed within the chamber, the flow control member being rotatable within the chamber between an open position and a closed position; an inlet conduit extending downwardly from the chamber, the inlet conduit including an inlet flow channel having an inlet; an outlet conduit extending upwardly from the chamber, the outlet conduit including an outlet flow channel having an outlet; and a stem operatively coupled to the flow control member such that rotation of the stem about an axis of rotation of the stem causes a corresponding rotation of the flow control member within the chamber about an axis of rotation of the flow control member, wherein the axis of rotation of the flow control member is coaxially aligned with the axis of rotation of the stem, and wherein the outlet of the outlet flow channel is located above the axis of rotation of the of the flow control member; a manifold coupled to the inlet conduit of the gas valve such that the inlet is in fluid communication with the manifold; a burner tube coupled to the outlet conduit of the gas valve such that the outlet is in fluid communication with the burner tube; and a control knob operatively coupled to the stem such that rotation of the control knob causes a corresponding rotation of the stem.

17. The grill of claim 16, wherein a gas train of the gas valve includes an S-shaped portion defined by the chamber and the outlet flow channel.

18. The grill of claim 17, wherein the outlet flow channel includes a first segment extending upwardly from and being in fluid communication with the chamber, a second segment extending forwardly from and being in fluid communication with the first segment, a third segment extending upwardly from and being in fluid communication with the second segment, and a fourth segment extending rearwardly from and being in fluid communication with the third segment, wherein the fourth segment includes the outlet, wherein the S-shaped portion of the gas train includes the chamber, the first segment, the second segment, the third segment, and the fourth segment.

19. The grill of claim 18, wherein the first segment, the second segment, the third segment, and the fourth segment are linear, wherein a central axis of the chamber, a central axis of the second segment, and a central axis of the fourth segment are parallel to one another and horizontally oriented, and wherein a central axis of the first segment and a central axis of the third segment are parallel to one another and vertically oriented.

20. The grill of claim 19, wherein the central axis of the second segment is located above the central axis of the chamber, the central axis of the fourth segment is located above the central axis of the second segment, the central axis of the third segment is located forward of the central axis of the first segment, and the central axis of the third segment is parallel to and located forward of a central axis of the inlet flow channel.
